

Unlocking Mobile & Remote Access: Conquering Expressway Certificates

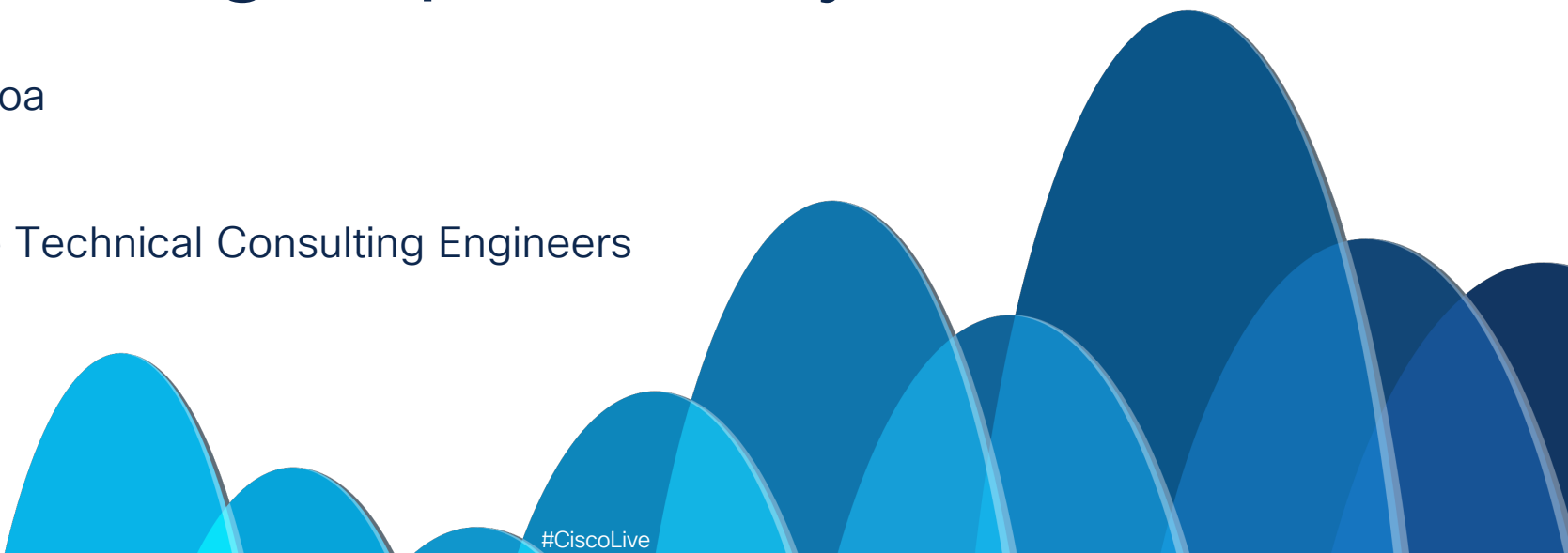
Jerson Gamboa
LT de Leon

Telepresence Technical Consulting Engineers

TACCOL-2011

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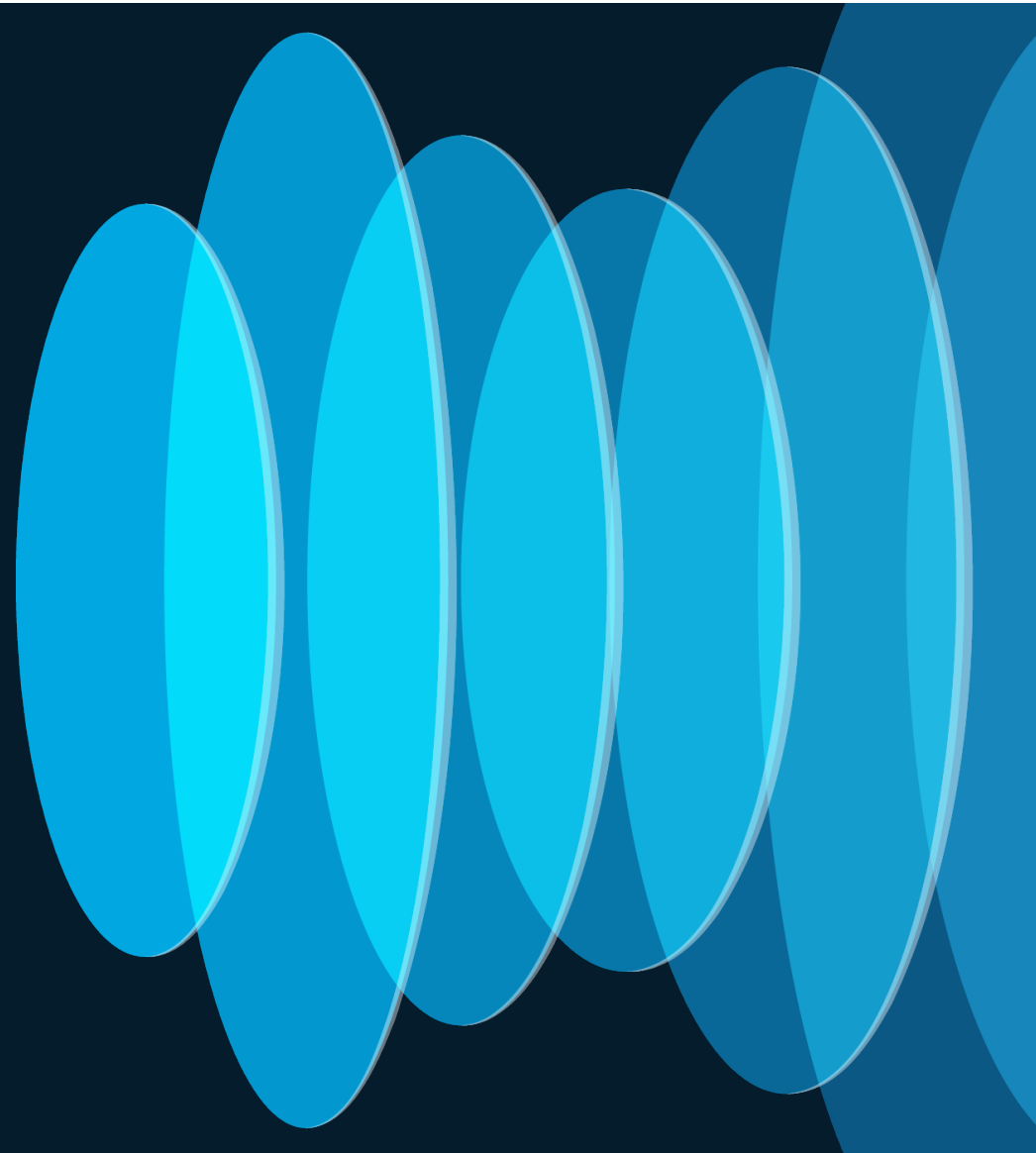


Agenda

- Introduction
- Reading Certificates
- MRA Certificates
- Troubleshooting Scenarios

Introduction

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Acronym Glossary Slide

- MRA = Mobile and Remote Access
- PKI = Public Key Infrastructure
- CSR = Certificate Signing Request
- CA = Certificate Authority
- CUCM = Cisco Unified Communications Manager
- TLS = Transport Layer Security

Public Key Infrastructure and Expressways



Private key: Used to decrypt the data sent to a server. (Decrypt CSR)



Signatures of data:
The file received from the CA is checked against the existing private key.



Certificate: X.509 Base64 encoded (PEM, CER, CRT for file extension)



Certificate chain: Checks the certificate received against the Certificate Authority.

Certificate Types



Default certificates are already installed on Expressway no changes are required. (It shows a temporary authority as the CA) and this is what you get when the server is a brand-new install.

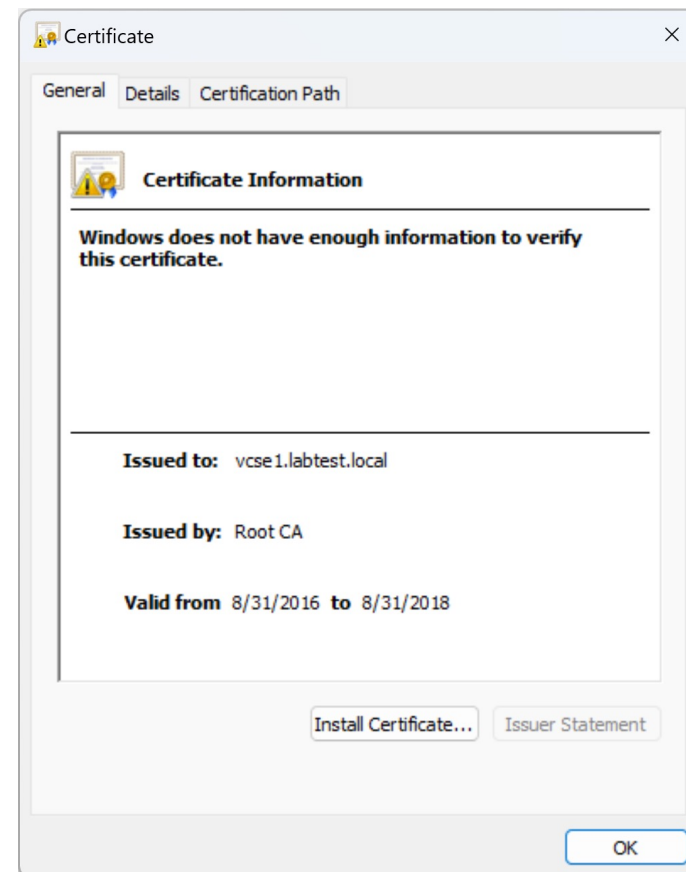
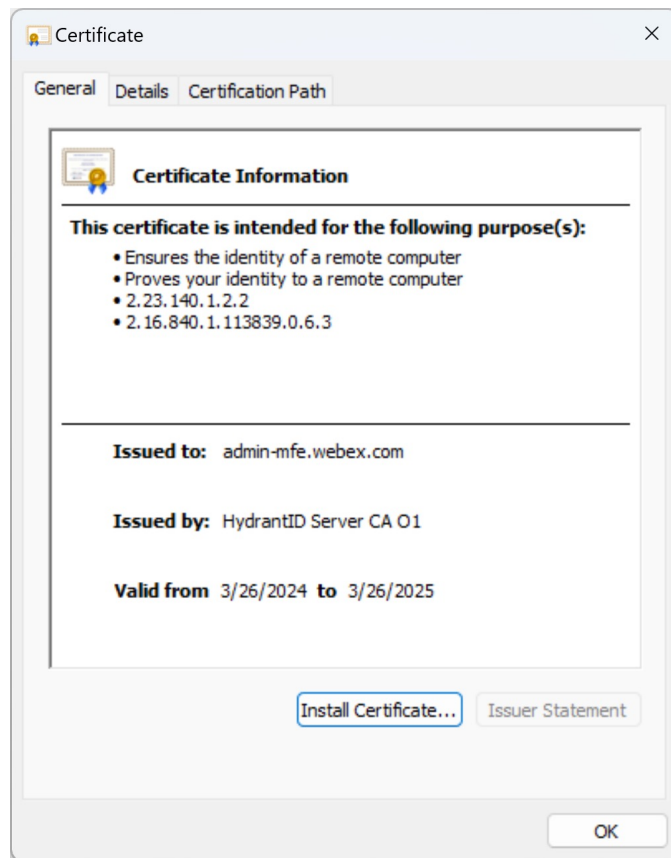


Self signed certificates, there are usually created within the Expressway using OpenSSL. Not supported by TAC, however it works fine in lab environments.



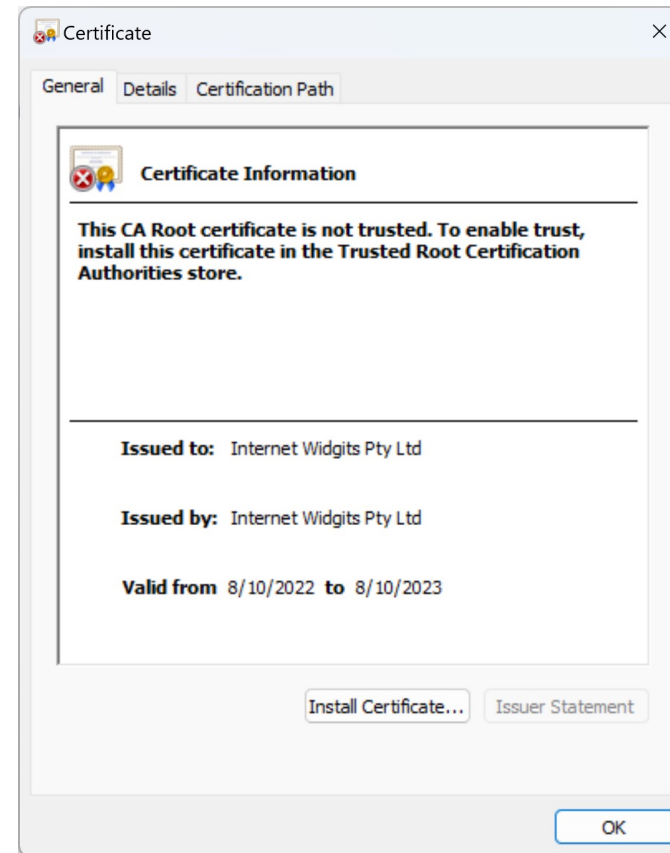
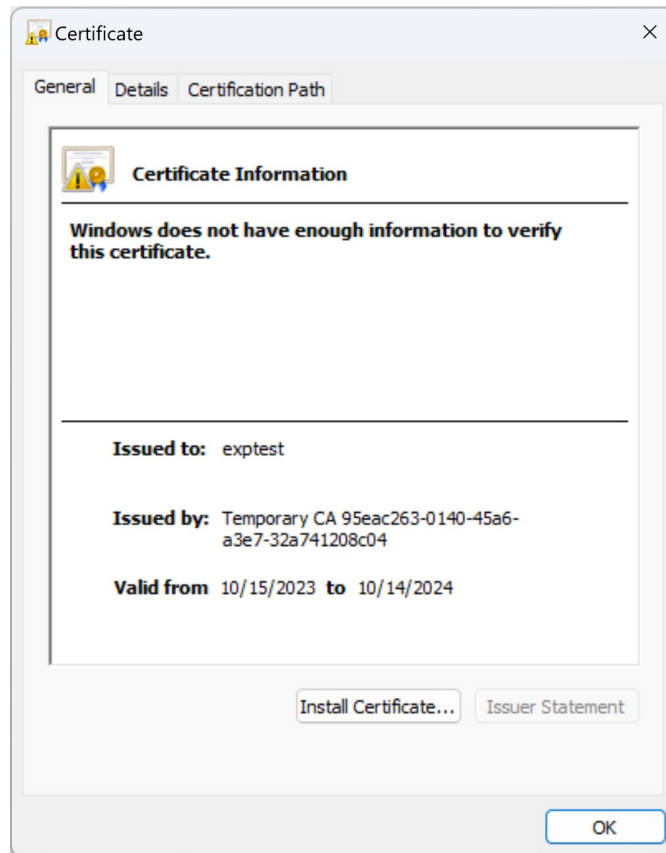
3rd party certificates will include a certificate chain with Root CA and Intermediate CA. This is the case for internal CA's (Like a Windows Server with CS Role) or public CA's (Like GoDaddy, DigiCert, etc.)

CA Signed Examples (supported)



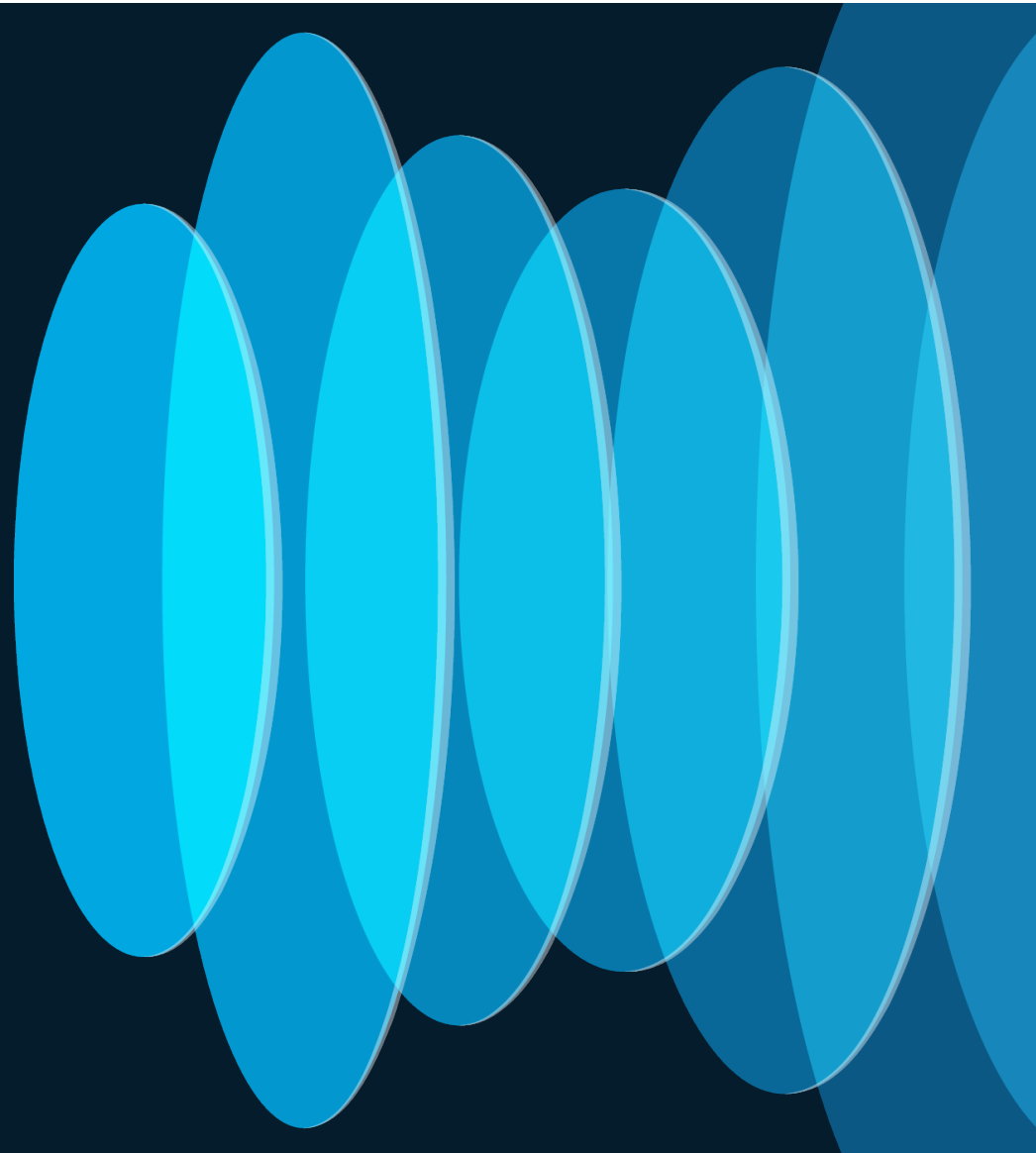
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Temporary and Self-Signed Examples (unsupported)

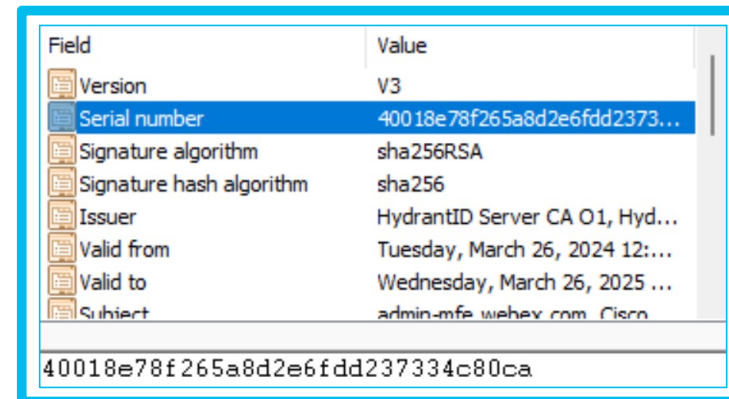
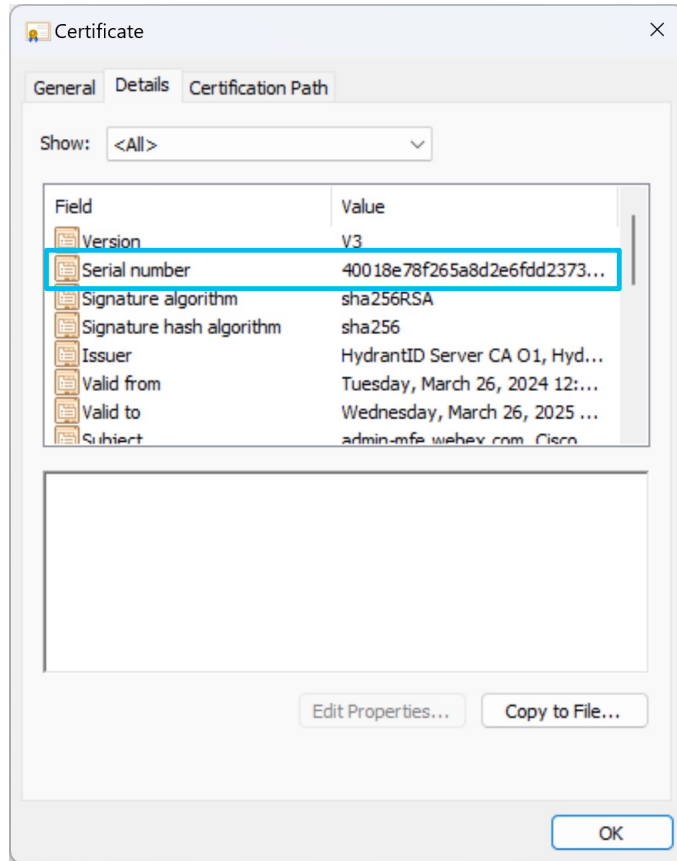


Reading Certificates

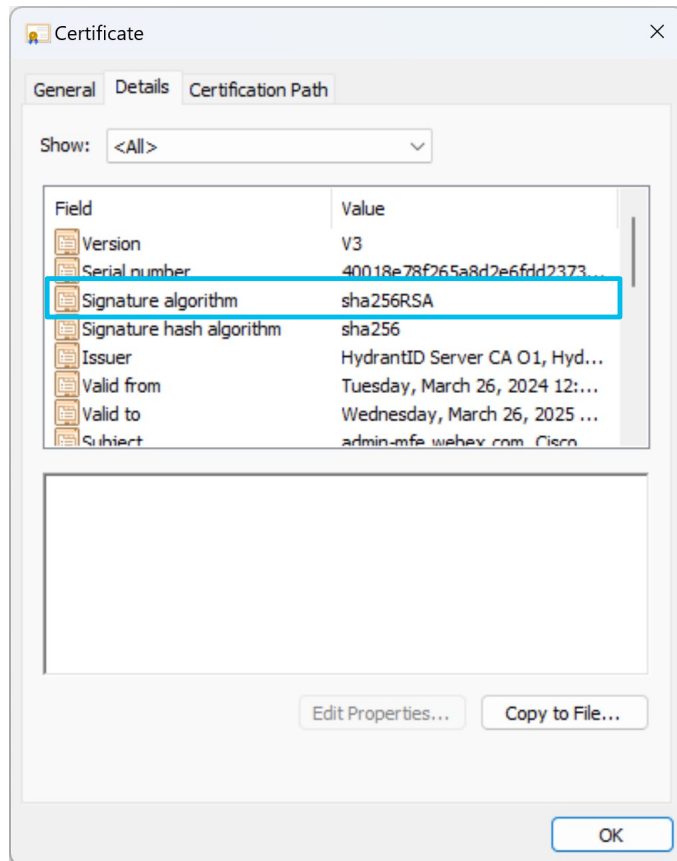
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What's in a certificate? – Serial Number



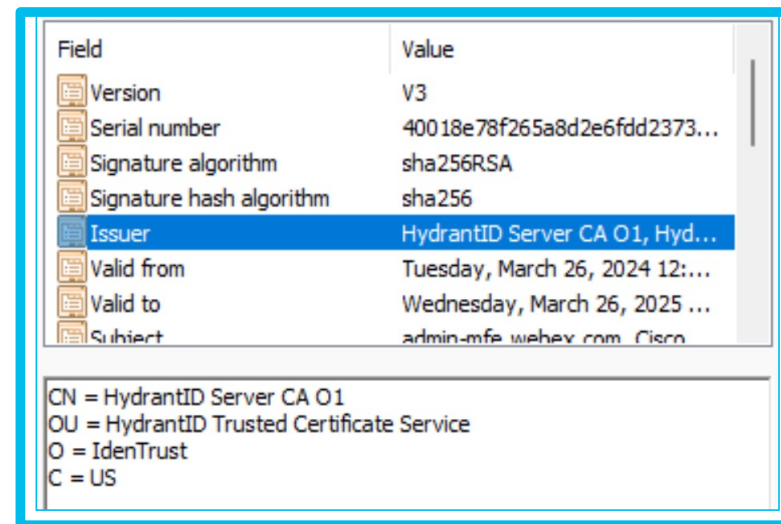
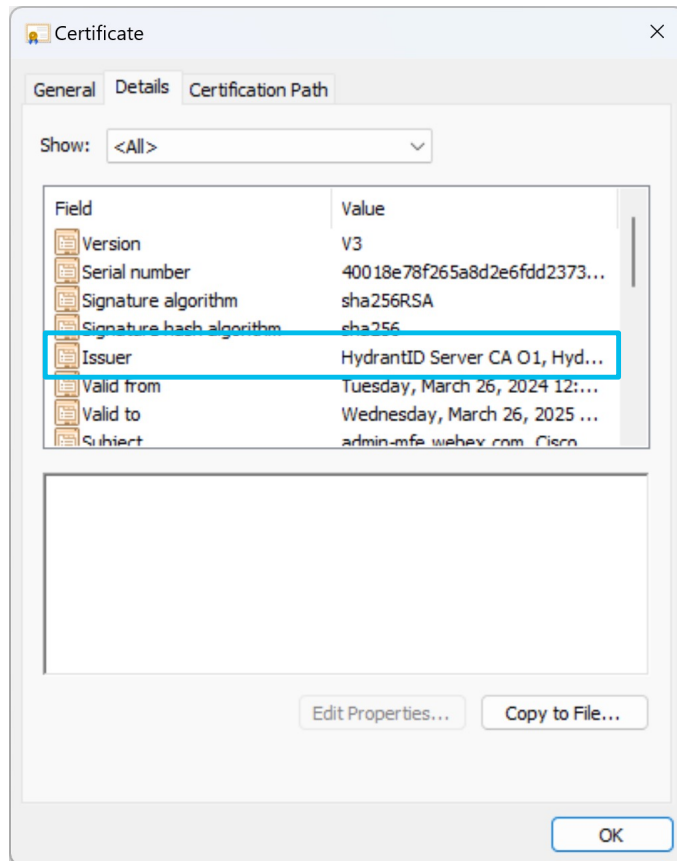
What's in a certificate? – Signature Algorithm



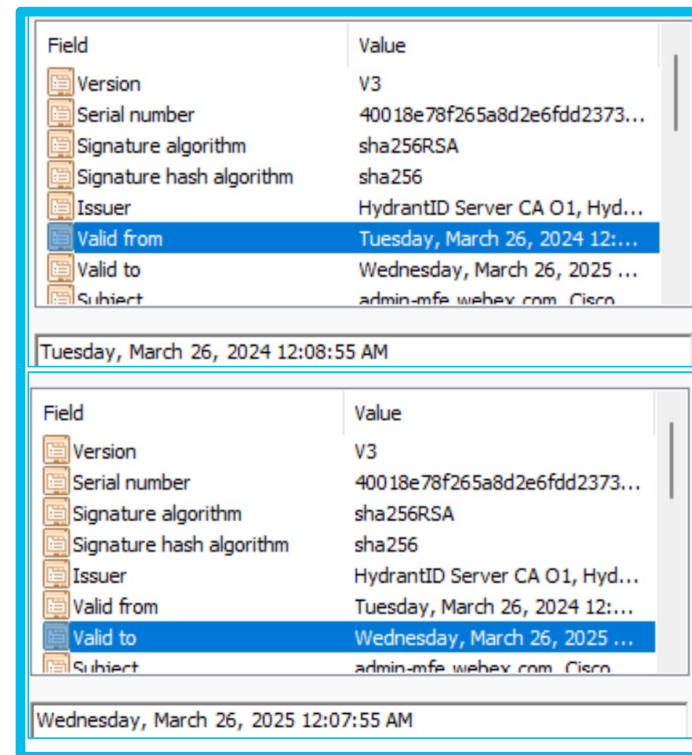
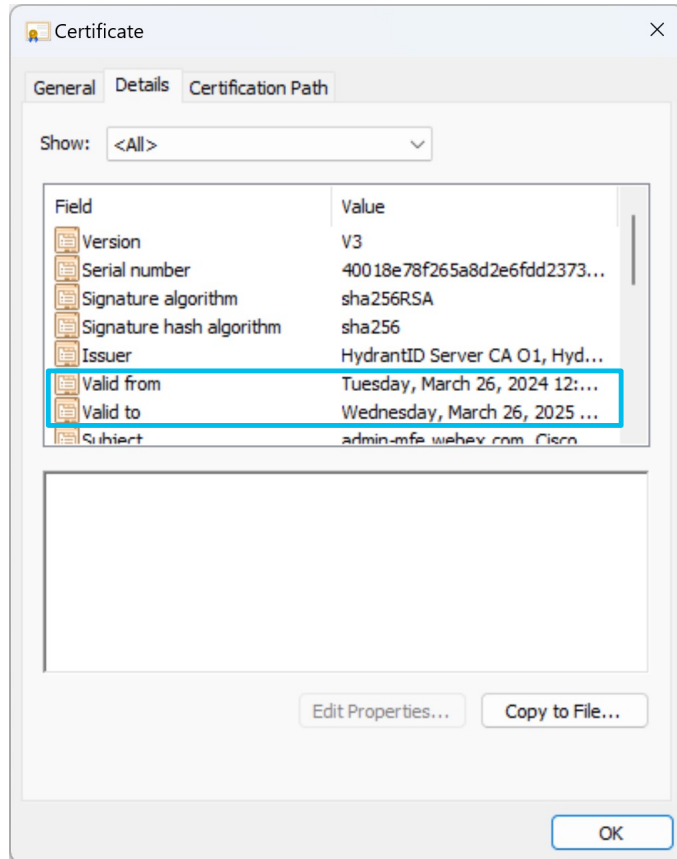
Field	Value
Version	V3
Serial number	40018e78f265a8d2e6fdd2373...
Signature algorithm	sha256RSA
Signature hash algorithm	sha256
Issuer	HydrantID Server CA O1, Hyd...
Valid from	Tuesday, March 26, 2024 12:...
Valid to	Wednesday, March 26, 2025 ...
Subject	admin-mfe.webex.com Cisco

sha256RSA

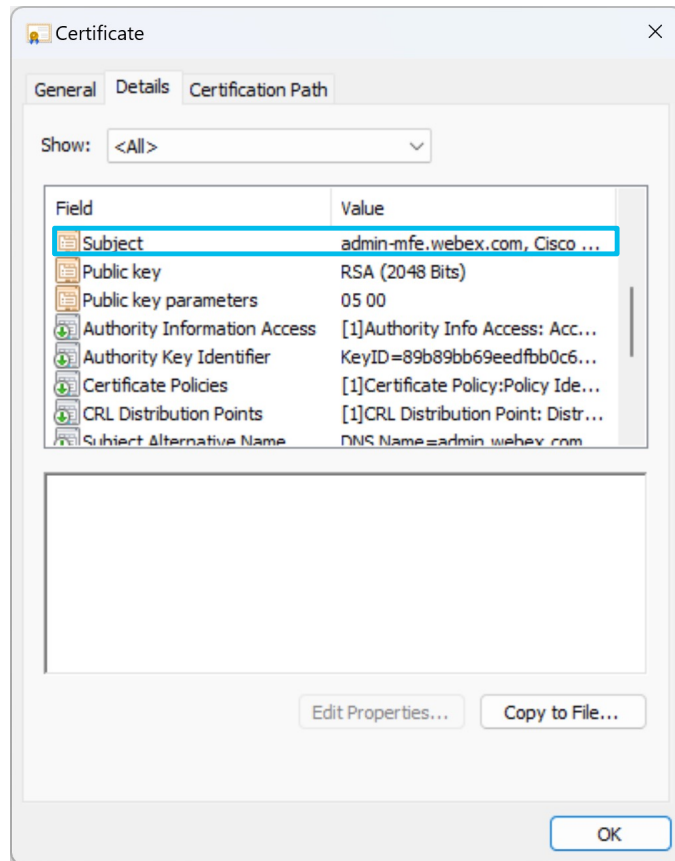
What's in a certificate? – Issuer



What's in a certificate? – Valid From/To



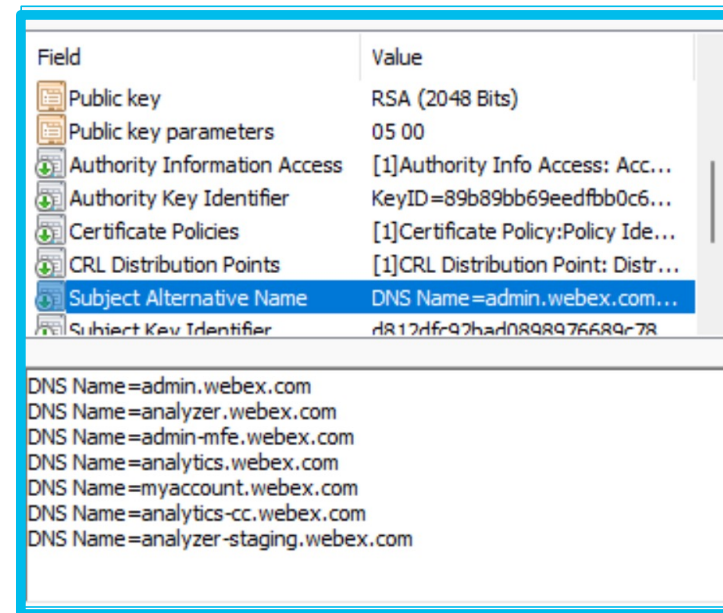
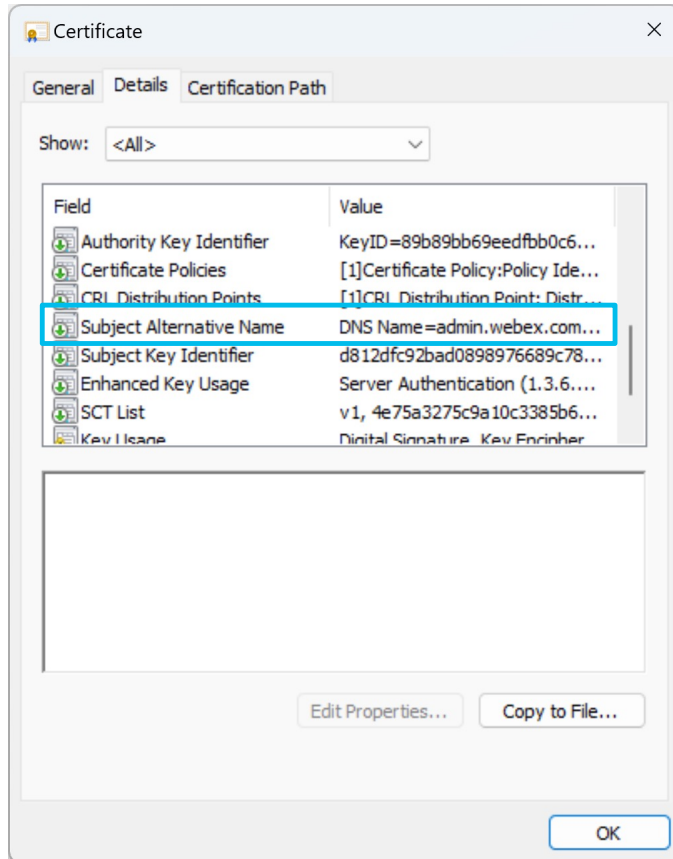
What's in a certificate? – Subject



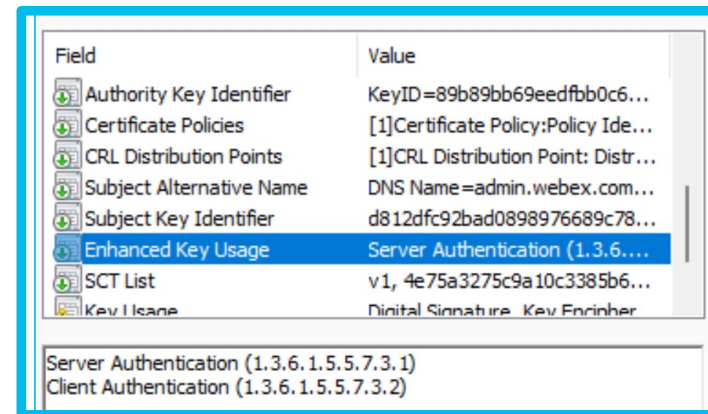
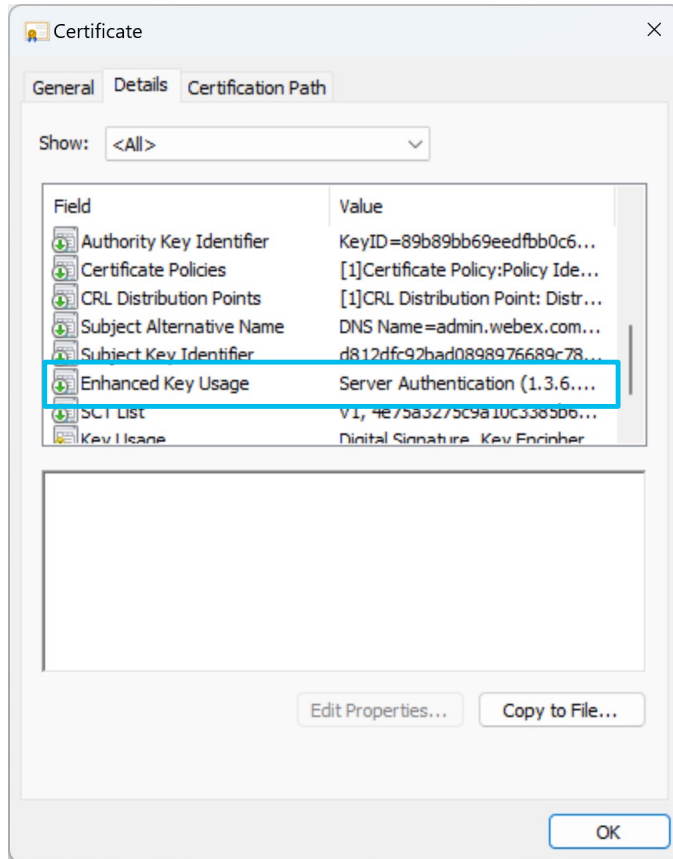
Field	Value
Subject	admin-mfe.webex.com, Cisco ...
Public key	RSA (2048 Bits)
Public key parameters	05 00
Authority Information Access	[1]Authority Info Access: Acc...
Authority Key Identifier	KeyID=89b89bb69eedfbb0c6...
Certificate Policies	[1]Certificate Policy:Policy Ide...
CRL Distribution Points	[1]CRL Distribution Point: Distr...
Subject Alternative Name	DNS Name=admin.webex.com

CN = admin-mfe.webex.com
O = Cisco Systems Inc.
L = San Jose
S = California
C = US

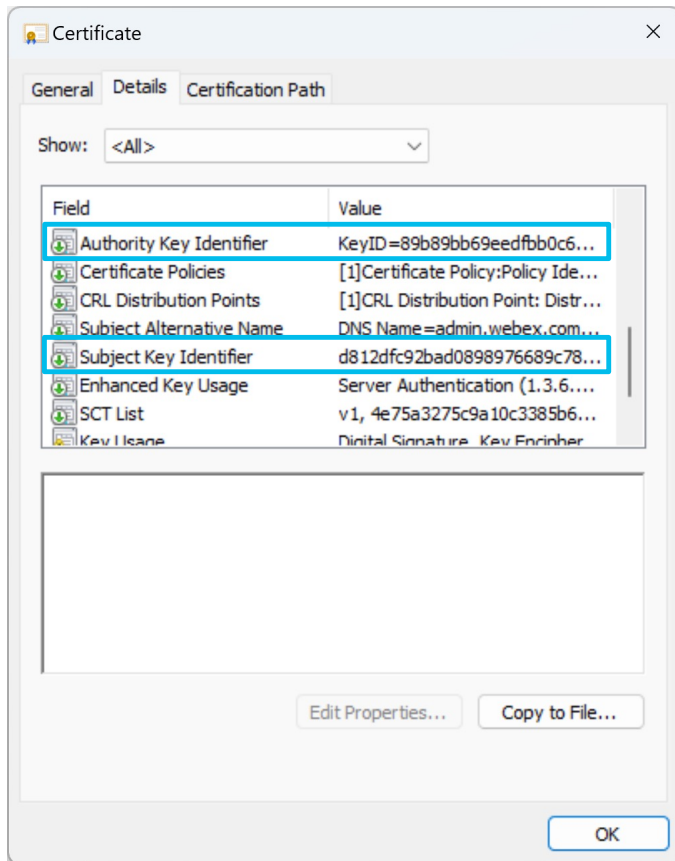
What's in a certificate? – Subject Alternative Name



What's in a certificate? – Enhanced Key Usage



What's in a certificate? – Authority/Subject Key Identifier



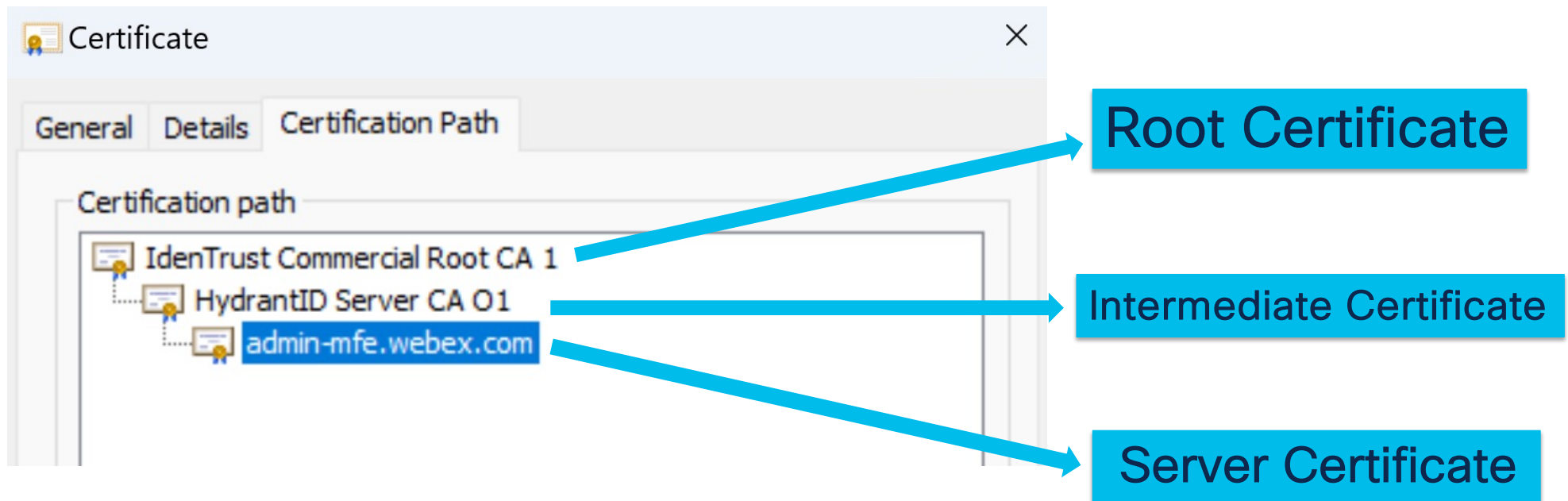
Field	Value
Authority Key Identifier	KeyID=89b89bb69eedfbb0c6...
Certificate Policies	[1]Certificate Policy:Policy Ide...
CRL Distribution Points	[1]CRL Distribution Point: Distr...
Subject Alternative Name	DNS Name=admin.webex.com...
Subject Key Identifier	d812dfc92bad0898976689c78...
Enhanced Key Usage	Server Authentication (1.3.6....
SCT List	v1, 4e75a3275c9a10c3385b6...
Key Usage	Digital Signature, Key Encipher...

KeyID=89b89bb69eedfbb0c6bd0dec674e3ca3929d2df9

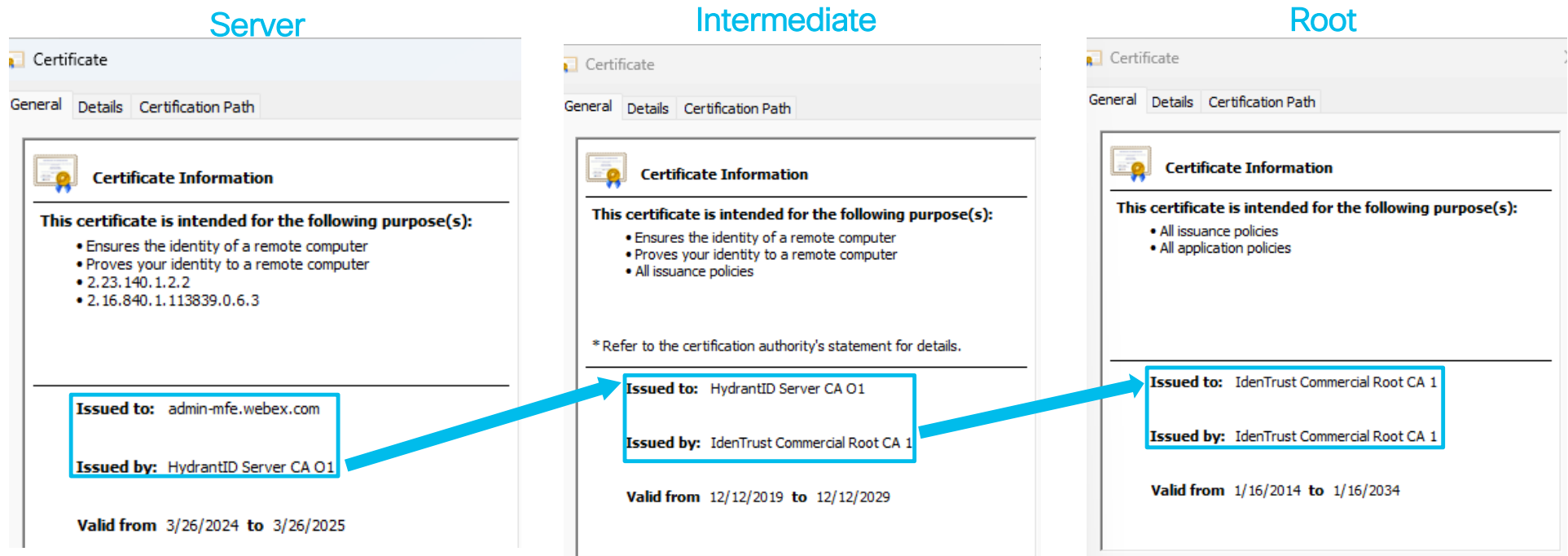
Field	Value
Authority Key Identifier	KeyID=89b89bb69eedfbb0c6...
Certificate Policies	[1]Certificate Policy:Policy Ide...
CRL Distribution Points	[1]CRL Distribution Point: Distr...
Subject Alternative Name	DNS Name=admin.webex.com...
Subject Key Identifier	d812dfc92bad0898976689c78...
Enhanced Key Usage	Server Authentication (1.3.6....
SCT List	v1, 4e75a3275c9a10c3385b6...
Key Usage	Digital Signature, Key Encipher...

d812dfc92bad0898976689c7846307d3a4fb2b73

What's in a certificate? - Certificate Chain



What's in a certificate? – Chain verification



What's in a certificate? – Chain verification

Certificate Server

General Details Certification Path

Show: <All>

Field	Value
Authority Information Access	[1]Authority Info Access: Acc...
Authority Key Identifier	KeyID=89b89bb69eedfbb0c6...
Certificate Policies	[1]Certificate Policy:Policy Ide...
CRL Distribution Points	[1]CRL Distribution Point: Distr...
Subject Alternative Name	DNS Name=admin.webex.com...
Subject Key Identifier	d812dfc92bad0898976689c78...
Enhanced Key Usage	Server Authentication (1.3.6....
SCT List	v1 4e75a3275c9a10c3385b6

KeyID=89b89bb69eedfbb0c6bd0dec674e3ca3929d2df9

Certificate Intermediate

General Details Certification Path

Show: <All>

Field	Value
Public key parameters	05 00
Authority Information Access	[1]Authority Info Access: Acc...
Authority Key Identifier	KeyID=ed4419c0d3f0068bee...
Certificate Policies	[1]Certificate Policy:Policy Ide...
CRL Distribution Points	[1]CRL Distribution Point: Distr...
Subject Key Identifier	89b89bb69eedfbb0c6bd0dec6...
Enhanced Key Usage	Server Authentication (1.3.6....
Basic Constraints	Subject Type=CA Path Length

89b89bb69eedfbb0c6bd0dec674e3ca3929d2df9

What's in a certificate? – Chain verification

The image displays two side-by-side screenshots of the Windows Certificate Manager application, illustrating the relationship between an Intermediate certificate and its Root certificate during chain verification.

Left Window: Intermediate Certificate

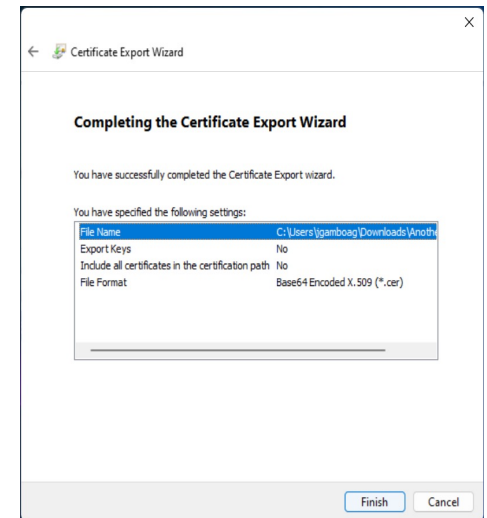
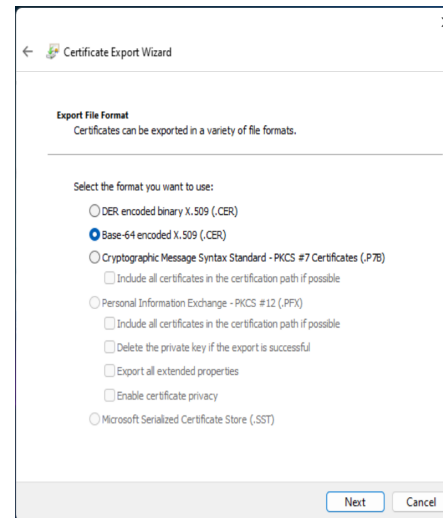
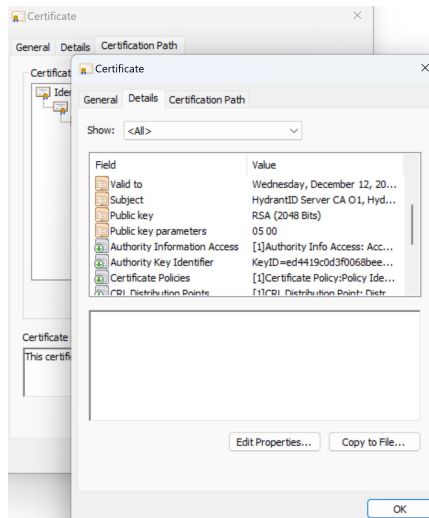
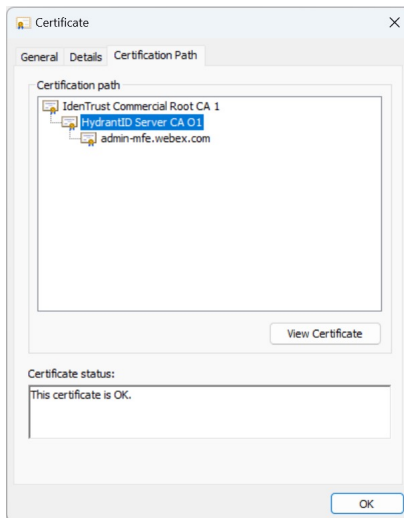
- Field:** Authority Key Identifier
- Value:** KeyID=ed4419c0d3f0068bee...

Right Window: Root Certificate

- Field:** Subject Key Identifier
- Value:** ed4419c0d3f0068beea47bbe4...

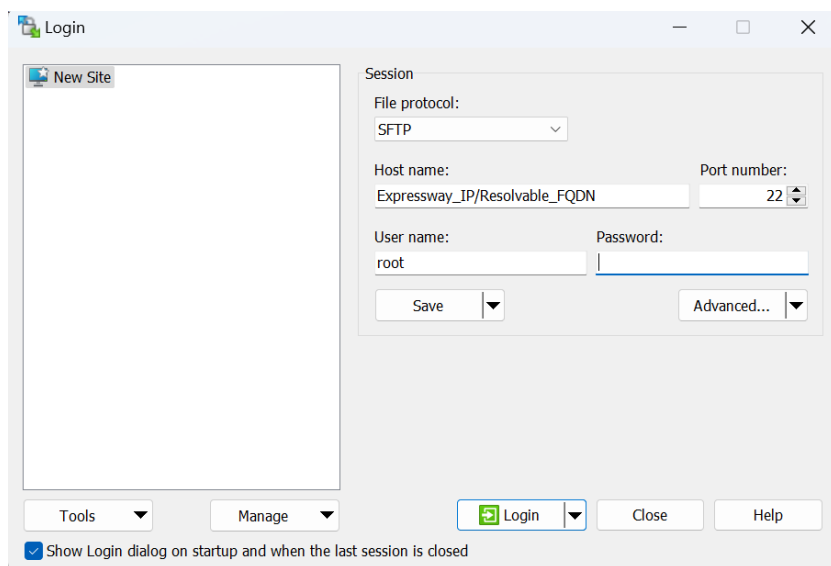
Both windows show the **KeyID** field at the bottom, which contains the hexadecimal value: `ed4419c0d3f0068beea47bbe42e72654c88e3676`. This value is highlighted in blue in both windows, demonstrating that the Authority Key Identifier of the Intermediate certificate matches the Subject Key Identifier of the Root certificate, confirming the chain of trust.

Extracting Certificates – Extract from Certificate



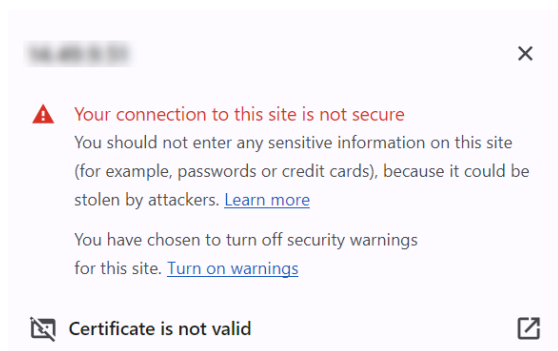
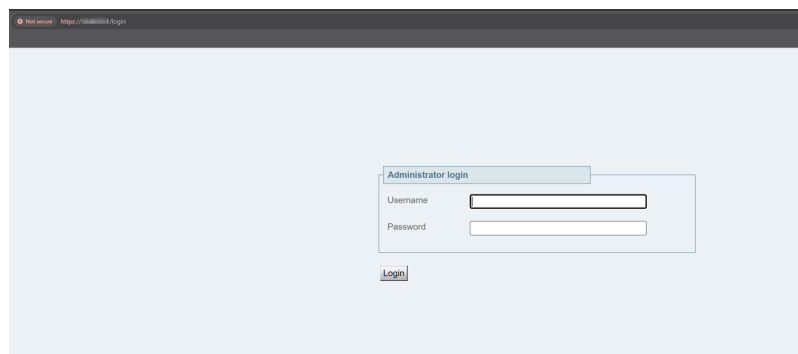
Extracting Certificates - WinSCP

Navigate to Tandberg/Persistent/Certs, collect certificates as required



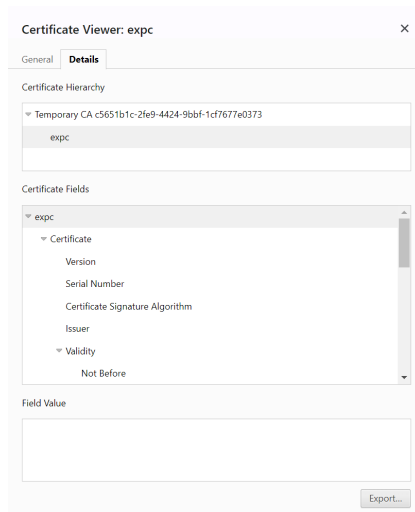
/tandberg/persistent/certs/				
Name	Size	Changed	Rights	Owner
.		4/29/2024 1:22:59 PM	rwxr-xr-x	root
saml		12/6/2022 12:58:43 PM	rwxr-xr-x	_samlvfy
multidomaincerts		3/5/2023 6:40:54 AM	rw-----	root
generated_csr		12/6/2022 12:54:43 PM	rwxr-xr-x	_nobody
mtls_ca.pem	9 KB	7/14/2021 7:05:57 AM	rw-r--r--	_nobody
sch_server_cert.pem	6 KB	3/13/2023 10:47:12 AM	rw-r--r--	_nobody
ca.pem	6 KB	4/24/2023 7:24:06 AM	rw-r--r--	_nobody
server-ssh.pem	4 KB	3/11/2024 9:21:55 AM	rw-----	_pfwd
privkey.pem.default	2 KB	6/7/2023 9:24:01 AM	r-----	root
privkey.pem	2 KB	4/23/2023 4:27:00 PM	r--w----	root
server.pem.default	2 KB	6/7/2023 9:24:01 AM	r--r--r--	root
server.pem	2 KB	4/23/2023 4:27:00 PM	rw-r--r--	_nobody
ca.pem.default	2 KB	6/7/2023 9:24:01 AM	r--r--r--	root
client-ca.crl.default	2 KB	4/24/2023 7:22:13 AM	rw-r--r--	root
client-ca.crl	2 KB	4/24/2023 7:22:13 AM	rw-r--r--	_nobody
policy-services.crl.default	2 KB	7/14/2021 7:05:57 AM	rw-r--r--	root
policy-services.crl	2 KB	7/14/2021 7:05:57 AM	rw-r--r--	_nobody
crl-update.conf	1 KB	3/11/2024 9:21:54 AM	rw-r--r--	root

Extracting Certificates – Web Interface



1. Navigate to the expressway web interface
2. Click on the padlock icon next to the URL
3. Select the certificate option

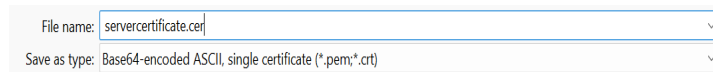
Extracting Certificates – Web Interface



4. While on details, select the server certificate

5. Click export

6. Browse and download the file on your desired folder

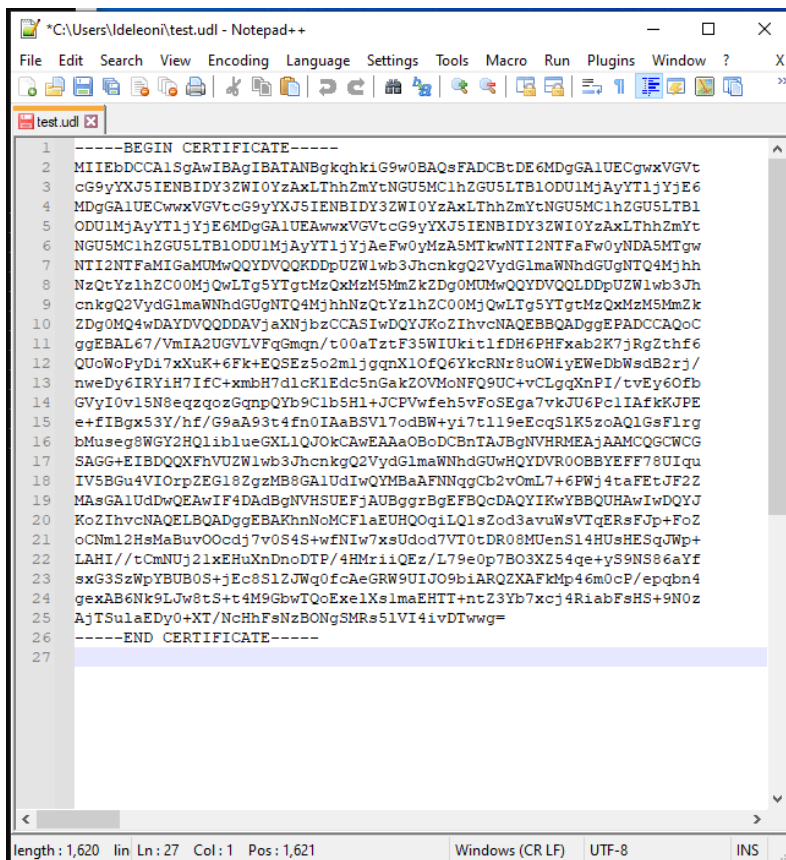


Extracting Certificates – Command Line (CLI)

[illegible]

1. Log into the Expressway CLI with admin credentials
2. Run the command 'xCommand Certs
Command: cert_get'
3. Copy the certificate output from ----
BEGIN CERTIFICATE----- through -----
END CERTIFICATE-----

Extracting Certificates – Command Line (CLI)



```
1 -----BEGIN CERTIFICATE-----
2 MIIEbDCCA1SgAwIBAgIBATANBgkqhkiG9w0BAQsFADCBtDE6MDgGA1UECgwvVGVT
3 cG9yYXJ5IENBIDY3ZWl0YzAxLThhZmYtNGU5MCIhZGU5LTB1ODU1MjAyYTljYjE6
4 MDgGA1UECwwvVGVTcG9yYXJ5IENBIDY3ZWl0YzAxLThhZmYtNGU5MCIhZGU5LTB1
5 ODU1MjAyYTljYjE6MDgGA1UEAwvVGVTcG9yYXJ5IENBIDY3ZWl0YzAxLThhZmYt
6 NGU5MCIhZGU5LTB1ODU1MjAyYTljYjE6MDgGA1UECgwvVGVTcG9yYXJ5IENBIDY3
7 NTI2NTFhMIGAMUMwQYDVQQKDDpUZlwlw3JhcnkgQ2VydG1maWNhdGUgNTQ4Mjhh
8 NzQtYzlhZC00MjQwLTg5YTgtMzQxMzMsMmZkZDg0MUMwQYDVQQLEDPUZlwlw3Jh
9 cnkgQ2VydG1maWNhdGUgNTQ4MjhhNzQtYzlhZC00MjQwLTg5YTgtMzQxMzMsMmZk
10 ZDg0MUMwQYDVQDQDAVjaXNjbzCCASiDQYJKoZIhvcNAQEBBQADggEPADCCAQoC
11 ggEBAL67/VmIA2UGVLFVqGmqn/t00aTztF35WUkIt1fDH6PHFxab2K7jRgZthf6
12 QUoWoPyD17xXuK+6Fk+EQSEz5o2mljgqnx10FQ6YkcrNz8uOWiyEWEDbWsdB2rj/
13 nweDy6IRYiH7IfC+ymbH7d1cK1Edc5nGakZOVmNFQ9UC+vCLgqXnPI/tvEy6Ofb
14 GVyI0v15N8eqzqozGqmpQYb9C1b5H1+JCPVwfeh5vFoSEga7vkJU6Pc1IAfKkJPE
15 e+fIBgx53Y/hf/G9aA93t4fn0IAaBSV17odBW+yi7t119eEcqS1K5zoAQ1GsFlrg
16 bMuseg8WGY2HQ1blblueGXL1QJOkCAwEAAaOBDCBnTAJBgNVHRMEAjAAMCQGCGWC
17 SAGG+ElBDQXhFhVUZlwlw3JhcnkgQ2VydG1maWNhdGUwHQYDVRO0BBYEFF78UIqu
18 IVSBGu4VIOrpZEG18ZgzMB8GA1UdIwQYMBaAFNNqgCb2vOmL7+6FWj4taFETJF2Z
19 MAsGA1UdDwQEAwIF4DAdBgNVHSUEFjAUBggqrBgEFBQcDAQYIKwYBBQUHAWIwDQYJ
20 KoZIhvcNAQELBQADggEBAKhnNoMCF1aEUHQOqiLQ1sZod3avuWsvTqERsFJp+FoZ
21 oCnm12HsMaBuvOocdj7v0S4S+wfNIw7xsUdod7VT0tDR08MUenS14HUsHESqJWp+
22 LAHI//tCmNUj21xEHuXnDnoDTP/4HMriiQEz/L79e0p7B03XZ54qe+yS9NS86aYf
23 sxG3SszWpYUB0S+jEc8S1ZJWq0fcAeGRW9UIJO9biARQZXAfkMp46m0cP/epqbn4
24 gexAB6Nk9LJw8tS+t4M9GbwTQoExelXslmaEHTT+ntZ3Yb7xcj4RiabFshS+9N0z
25 AjTSulaEDy0+XT/NcHhFsnZBONGSMRs51VI4ivDTIwgm=
26 -----END CERTIFICATE-----
27
```

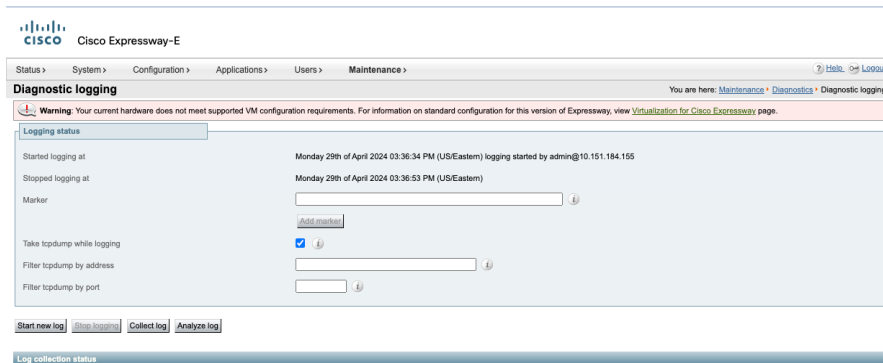
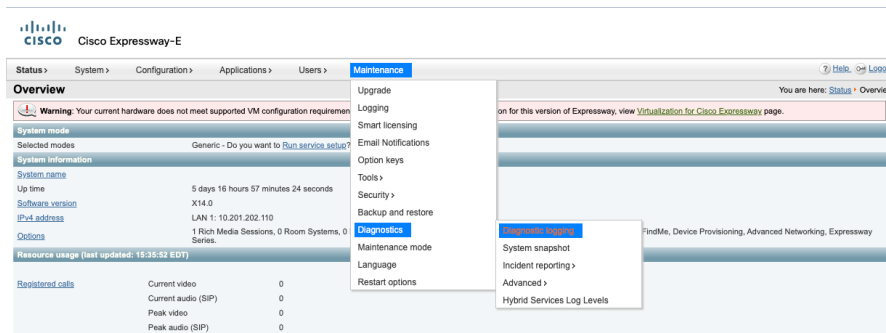
4. Paste the certificate into a notepad application like Notepad++

5. Make sure that there is a carriage return after -----END CERTIFICATE-----

6. Save file as <certname>.pem*

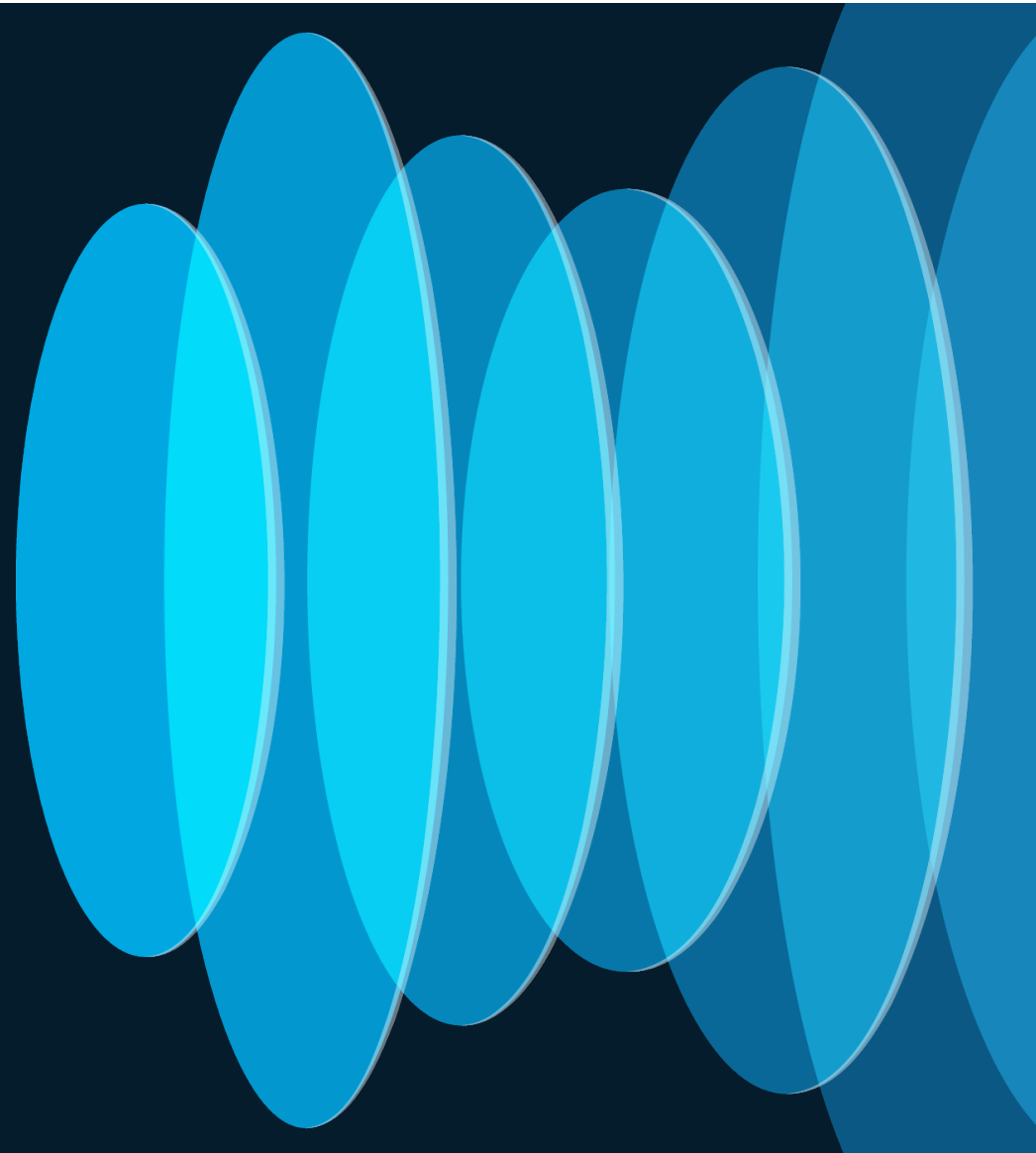
* When saving make sure the Save as type is set to All types (*.*)

Extracting Certificates – Log Bundle

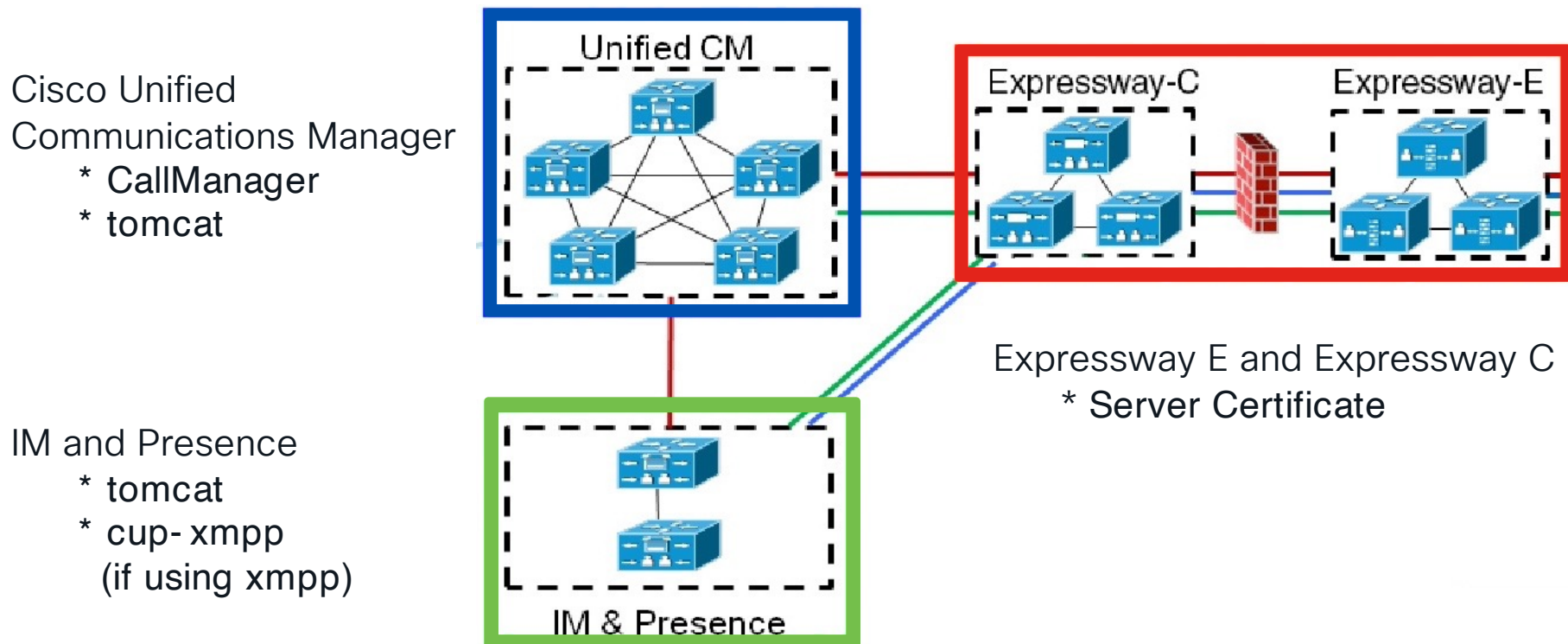


1. Log into the web UI
2. Navigate to Maintenance > Diagnostics > Diagnostic Logging
3. Click Start new log > Stop logging > Collect log > Download log
4. The server .pem certificate file will be included in the tar.gz log bundle

MRA Certificates



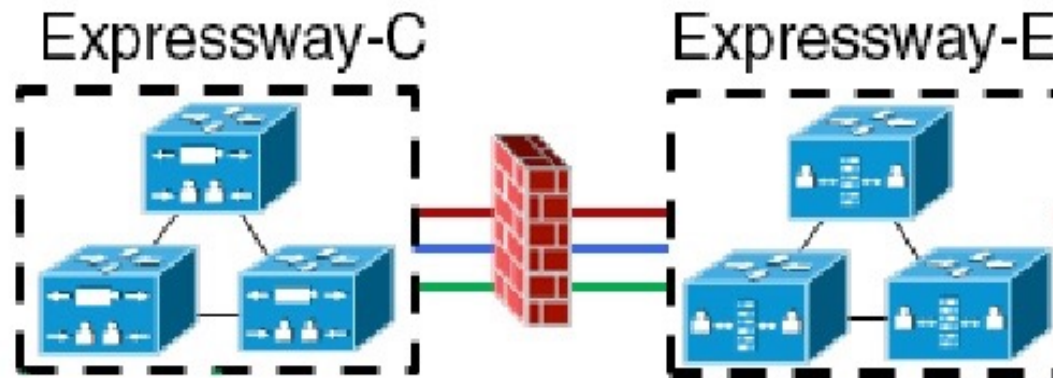
Significant Certificates in an MRA Deployment



Expressway SAN Requirements

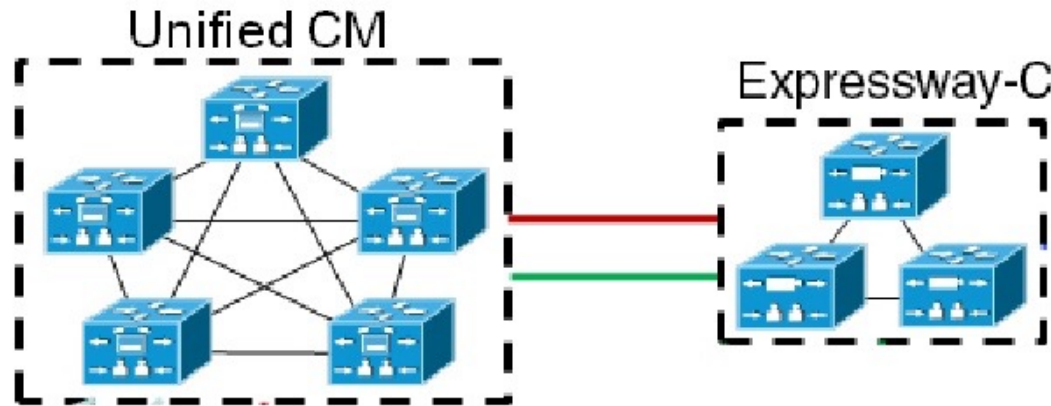
Add these items as subject alternative names	When generating a CSR for these purposes			
	Mobile and Remote Access	Jabber Guest	XMPP Federation	Business to Business Calls
Unified CM registrations domains (despite their name, these have more in common with service discovery domains than with Unified CM SIP registration domains)	Required on Expressway-E only	–	–	–
XMPP federation domains	–	–	Required on Expressway-E only	–
IM and Presence chat node aliases (federated group chat)	–	–	Required	–
Unified CM phone security profile names	Required on Expressway-C only	–	–	–
(Clustered systems only) Expressway Cluster name	Required on Expressway-C only	Required on Expressway-C only	Required on Expressway-C only	–

UC Certificate Exchange: Establishing Trust



- Expressway C signing CA certificate installed in Expressway E Trusted CA list
- Expressway E signing CA certificate installed in Expressway C Trusted CA list

UC Certificate Exchange: Establishing Trust



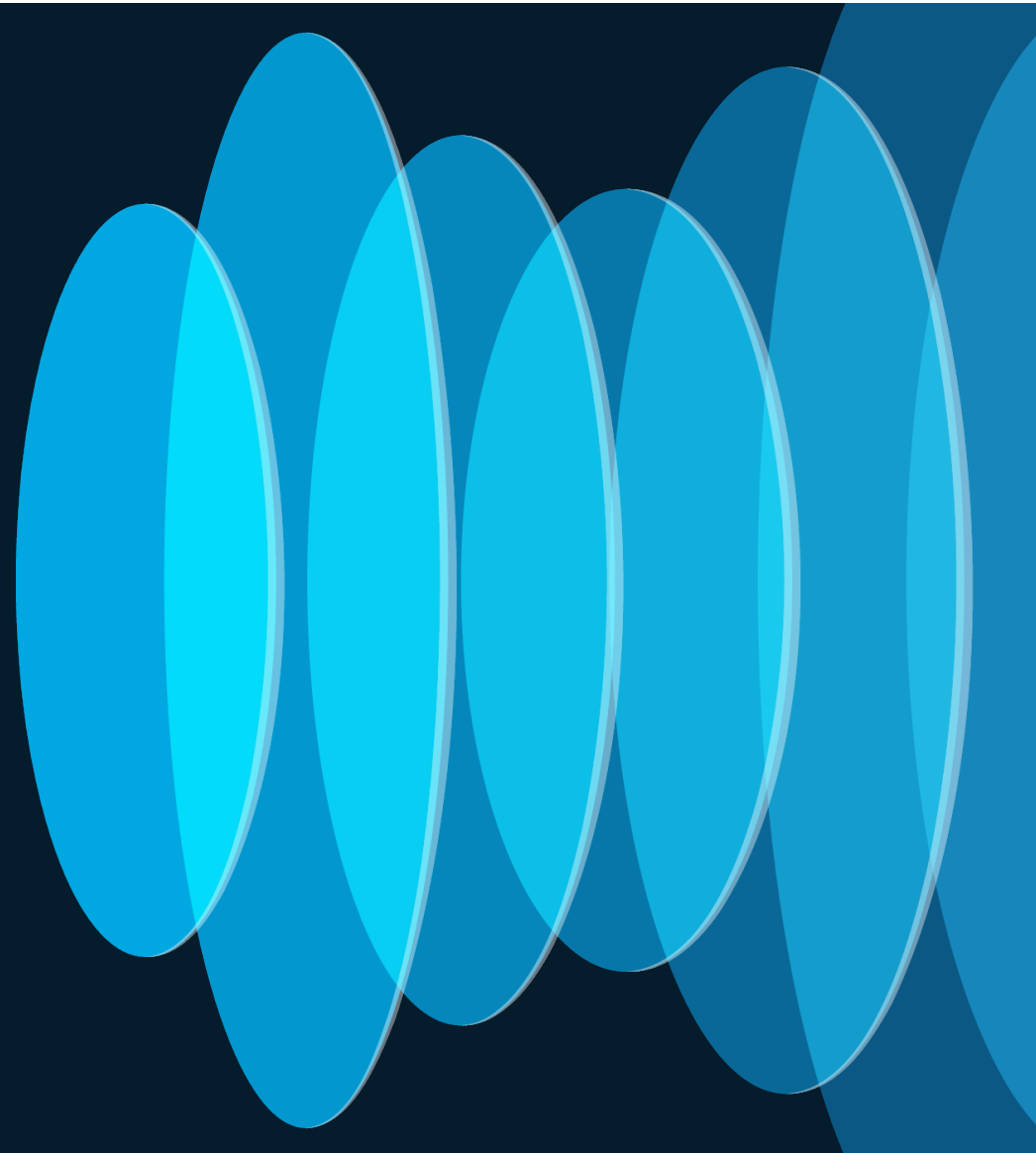
- Expressway C signing CA certificate installed in CUCM tomcat-trust and CallManager-trust
- CallManager and tomcat signing CA certificate installed in Expressway C Trusted CA list

*CA signed certificates are recommended.

*Expressway only recognizes one self-signed certificate per Common Name so only one of Tomcat or CallManager can be self-signed.

Troubleshooting Scenarios

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TLS handshake failures and verification

TCP Handshake

TCP	66	58563 → 443	[SYN]	Seq=0	Win=64240	Len=0	MSS=1460	WS=256	SACK_PERM
TCP	66	443 → 58563	[SYN, ACK]	Seq=0	Ack=1	Win=64240	Len=0	MSS=1460	SACK_PERM WS=128
TCP	60	58563 → 443	[ACK]	Seq=1	Ack=1	Win=262656	Len=0		

TLSv1.2	837	Client Hello
TCP	54	443 → 58563 [ACK] Seq=1 Ack=784 Win=64128 Len=0
TLSv1.2	210	Server Hello, Change Cipher Spec, Encrypted Handshake Message
TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
TCP	54	443 → 58563 [ACK] Seq=157 Ack=835 Win=64128 Len=0
TLSv1.2	912	Application Data

TLS Handshake

TLS handshake failures and verification

TCP Handshake

admin	expressway-c	TCP	66 58559 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
expressway-c	admin	TCP	66 443 → 58559 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
admin	expressway-c	TCP	60 58559 → 443 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
admin	expressway-c	TLSv1.2	741 Client Hello
expressway-c	admin	TCP	54 443 → 58559 [ACK] Seq=1 Ack=688 Win=64128 Len=0
expressway-c	admin	TLSv1.2	210 Server Hello, Change Cipher Spec, Encrypted Handshake Message
admin	expressway-c	TLSv1.2	61 Alert (Level: Fatal, Description: Certificate Unknown)
expressway-c	admin	TCP	54 443 → 58559 [ACK] Seq=157 Ack=695 Win=64128 Len=0

TLS Handshake

Invalid Certificate

TLS handshake failures and verification

TCP Handshake

cucm	expressway-c	TCP	74 33662 → 6972 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3341487451 TSecr=0 WS=128
expressway-c	cucm	TCP	74 6972 → 33662 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=1382887028 TSecr=3341487451 WS=128
cucm	expressway-c	TCP	66 33662 → 6972 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3341487451 TSecr=1382887028
cucm	expressway-c	TLSv1.2	583 Client Hello
expressway-c	cucm	TLSv1.2	4034 Server Hello, Certificate, Server Key Exchange, Certificate Request, Server Hello Done
cucm	expressway-c	TCP	66 33662 → 6972 [ACK] Seq=518 Ack=3969 Win=62464 Len=0 TSval=3341487462 TSecr=1382887039
cucm	expressway-c	TLSv1.2	4289 Certificate
expressway-c	cucm	TCP	66 6972 → 33662 [ACK] Seq=3969 Ack=4741 Win=38528 Len=0 TSval=1382887039 TSecr=3341487462
expressway-c	cucm	TLSv1.2	73 Alert (Level: Fatal, Description: Unknown CA)
cucm	expressway-c	TCP	66 33662 → 6972 [ACK] Seq=4741 Ack=3976 Win=64128 Len=0 TSval=3341487463 TSecr=1382887039
expressway-c	cucm	TCP	66 6972 → 33662 [FIN, ACK] Seq=3976 Ack=4741 Win=38528 Len=0 TSval=1382887040 TSecr=3341487463

TLS Handshake

Invalid Certificate Authority

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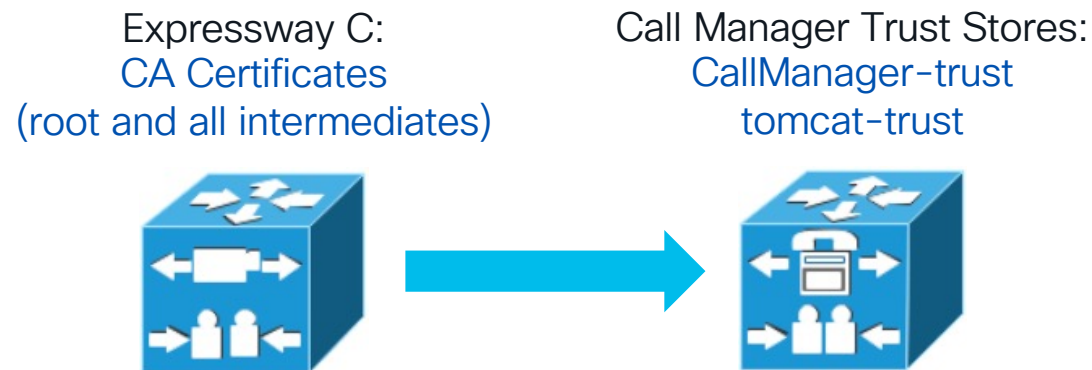
#CiscoLive TACCOL-2011

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Expressway C Traffic Server Enforces Certificate Verification

As of x14.0.2, Expressway will send its client certificate whenever a server (CUCM) requests it, for services running on ports other than 8443 (e.g. 6971,6972) even if CUCM is in non-secure mode.



Expressway C Traffic Server Enforces Certificate Verification

The identifying signature for this issue is the Expressway's connections to CUCM over **port 6972** failing with "**tlsv1 alert unknown ca**" and "**502 connect failed**" response.

In the client logs if you filter for 'HTTP response codes you will see **200 OK** responses to the initial HTTP messages and then a **502** when the client requests the CSF configuration file:

```
2023-12-12 18:05:24,011 INFO [0x00006d3c] [ls\src\http\BasicHttpClientImpl.cpp(675)] [csf.httpclient]
[csf::http::performRequest] - *-----* HTTP response code 200 connect code 0 for request #13 to
https://jabber.clus2024.com:8443/Y2x1czIwMjQuY29tL2g/get_edge_config?service_name=_cisco-uds
2023-12-12 18:05:24,359 INFO [0x00006d3c] [ls\src\http\BasicHttpClientImpl.cpp(675)] [csf.httpclient]
[csf::http::performRequest] - *-----* HTTP response code 200 connect code 0 for request #14 to
https://jabber.clus2024.com:8443/Y2x1czIwMjQuY29tL2h0dHBzL2NsdXNwdWIuY21zY28udGVzdC84NDQz/cucm-uds/version
2023-12-12 18:05:24,493 INFO [0x00006d3c] [ls\src\http\BasicHttpClientImpl.cpp(675)] [csf.httpclient]
[csf::http::performRequest] - *-----* HTTP response code 200 connect code 0 for request #16 to
https://jabber.clus2024.com:8443/Y2x1czIwMjQuY29tL2h0dHBzL2NsdXNwdWIuY21zY28udGVzdC84NDQz/cucm-uds/servers
2023-12-12 18:05:24,843 INFO [0x00006d3c] [ls\src\http\BasicHttpClientImpl.cpp(675)] [csf.httpclient]
[csf::http::performRequest] - *-----* HTTP response code 502 connect code 0 for request #19 to
https://jabber.clus2024.com:8443/Y2x1czIwMjQuY29tL2h0dHBzL2NsdXNwdWIuY21zY28udGVzdC82OTcy/CSFuser.cnf.xml
```

Expressway C Traffic Server Enforces Certificate Verification

Looking at the Expressway C logs we see the Expressway C sending the **CSFuser.cnf.xml** configuration request to the CUCM on **port 6972** used for TFTP:

```
2023-12-12T11:14:56.307-07:00 clus2024expc traffic_server[22698]: UTCTime="2023-12-12 18:14:56,307"
Module="network.http.trafficserver" Level="DEBUG": Detail="Sending Request" Txn-id="1410111"
TrackingID="8dlb73e1-cfee-4066-8432-be78ald1c000" Dst-ip="0.0.0.8" Dst-port="6972"
HTTPMSG:
|GET /CSFuser.cnf.xml HTTP/1.1
Host: 0.0.0.8:6972
Accept: */*
Cookie:<CONCEALED>
User-Agent: WebEx/1.0.0.2 (Windows;)
TrackingID: 8dlb73e1-cfee-4066-8432-be78ald1c000
Client-ip: 0.0.0.3
X-Forwarded-For: 0.0.0.3, 127.0.0.1
Via: https/1.1 vcs[793cc42c-a89c-4d85-bfac-abef6ee56000] (ATS), http/1.1 vcs[e00de37e-99fa-4ea8-aad9-
f92e1b306000] (ATS)
|
```

Expressway C Traffic Server Enforces Certificate Verification

We then see the SSL connection to the CUCM fail with the **tlsv1 alert unknown ca** failure followed by the **502 connect failed** response:

```
2023-12-12T11:14:56.345-07:00 clus2024expc traffic_server[22698]: [ET_NET 1] ERROR: SSL connection failed
for '0.0.0.8': error:14094418:SSL routines:ssl3_read_bytes:tlsv1 alert unknown ca

2023-12-12T11:14:56.382-07:00 clus2024expc traffic_server[22698]: UTCTime="2023-12-12 18:14:56,382"
Module="network.http.trafficserver" Level="DEBUG": Detail="Sending Response" Txn-id="1410111" TrackingID=""
Dst-ip="127.0.0.1" Dst-port="32794"
HTTPMSG:
|HTTP/1.1 502 connect failed
Date: Tue, 12 Dec 2023 18:14:56 GMT
Connection: keep-alive
Server: CE_C
Cache-Control: no-store
Content-Type: text/html
Content-Language: en
Content-Length: 247
|
```

Expressway C Traffic Server Enforces Certificate Verification

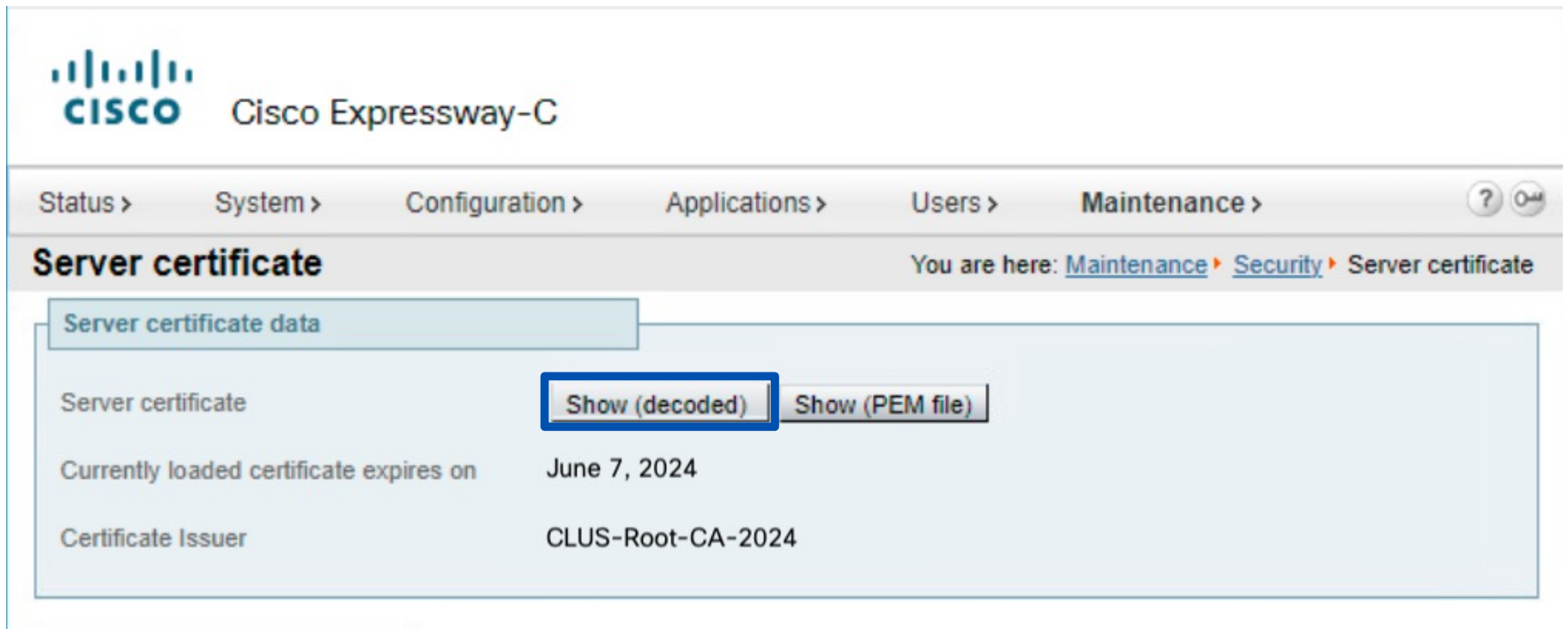
Confirm the serial number of the problematic certificate from the Expressway C packet capture.

4206	18:14:56.326271	Expressway C	CUCM	TCP	66	6972	35596 → 6972 [ACK] Seq=175 Ack=5963 Win=61440 Len=0
4236	18:14:56.344470	Expressway C	CUCM	TLSv1...	2786	6972	Certificate, Client Key Exchange, Certificate Verif
4238	18:14:56.345389	CUCM	Expressway C	TCP	66	35596	6972 → 35596 [ACK] Seq=5963 Ack=2895 Win=35840 Len=
4239	18:14:56.345582	CUCM	Expressway C	TLSv1...	73	35596	Alert (Level: Fatal, Description: Unknown CA)
4240	18:14:56.345588	Expressway C	CUCM	TCP	66	6972	35596 → 6972 [ACK] Seq=2895 Ack=5970 Win=64128 Len=
4241	18:14:56.345771	CUCM	Expressway C	TCP	66	35596	6972 → 35596 [FIN, ACK] Seq=5970 Ack=2895 Win=35840

> Ethernet II, Src: VMware_95: , Dst:
> Internet Protocol Version 4, Src: Expresswy C Dst: CUCM
> Transmission Control Protocol, Src Port: 35596, Dst Port: 6972, Seq: 175, Ack: 5963, Len: 2720
> Transport Layer Security
< TLSv1.2 Record Layer: Handshake Protocol: Certificate
Content Type: Handshake (22)
Version: TLS 1.2 (0x0303)
Length: 1996
< Handshake Protocol: Certificate
Handshake Type: Certificate (11)
Length: 1992
Certificates Length: 1989
< Certificates (1989 bytes)
Certificate Length: 1986
< Certificate [truncated]: <redacted for privacy>
< signedCertificate
version: v3 {2}
serialNumber: 0x531251g53485t416b4fb42e74b7ca6
signature (sha256WithRSAEncryption)
issuer: rdnSequence (0)
validity
< subject: rdnSequence (0)
rdnSequence: 4 items (id-at-commonName= expwyc.CLUS2024.com ,id-at-organizationName= Cisco Live 2024 ,id-at-stateOrProvince=
subjectPublicKeyInfo

Expressway C Traffic Server Enforces Certificate Verification

Check the installed Expressway C certificate and check the serial number to make sure it matches the serial number identified in the logs



The screenshot shows the Cisco Expressway-C web interface. The top navigation bar includes links for Status, System, Configuration, Applications, Users, and Maintenance. The 'Server certificate' page is active, with a breadcrumb trail: You are here: Maintenance > Security > Server certificate. A tab labeled 'Server certificate data' is selected. Below this, there are two buttons: 'Show (decoded)' and 'Show (PEM file)'. The 'Show (decoded)' button is highlighted with a blue box. Below the buttons, the following information is displayed:

Server certificate	
Currently loaded certificate expires on	June 7, 2024
Certificate Issuer	CLUS-Root-CA-2024

Expressway C Traffic Server Enforces Certificate Verification

Check the installed Expressway C certificate and check the serial number to make sure it matches the serial number identified in the logs

Certificate:
Data:
Version: 3 (0x2)
Serial Number:
0x:53:12:51:g5:34:85:t4:16:b4:fb:42:e7:4b:7c:a6
Signature Algorithm: sha256WithRSAEncryption
Issuer: DC=2011, DC=TACCOL, CN=CLUS-Root-CA-2024
Validity
Not Before: Jun 03 16:00:00 2024 GMT
Not After : Jun 07 00:00:00 2024 GMT
Subject: C=US, ST=NC, L=RTP, O=CISCO, OU=TP-LAB, CN=expwyc.CLUS2024.com
Subject Public Key Info:
Public Key Algorithm: rsaEncryption
Public-Key: (4096 bit)
Modulus:
00:92:2a:f4:82:f2:95:8d:c0:c8:ed:06:60:d0:45:
18:09:8f:99:d9:49:37:58:e4:74:16:33:28:8f:d3:
73:be:0f:b8:62:9f:46:30:0f:26:09:b3:6a:6b:93:
5d:92:62:24:06:ba:43:9e:5e:82:f2:ce:f5:39:3b:
e0:4c:e1:dd:a8:68:ca:56:21:37:96:8a:59:cf:c5:

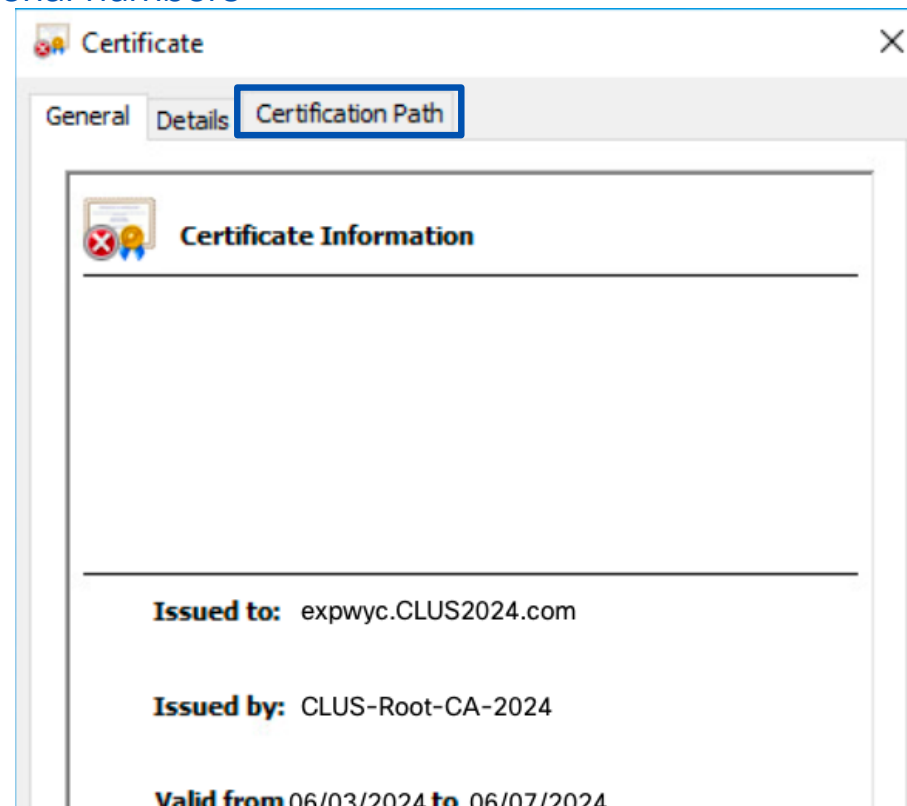
?

0

curity Server certificate

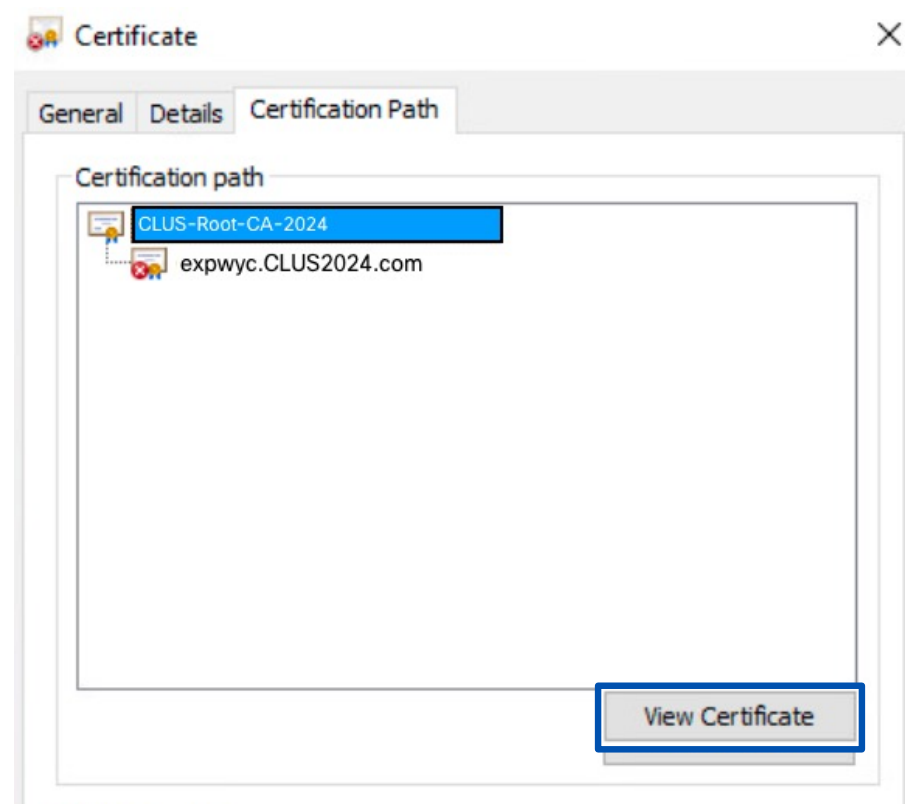
Expressway C Traffic Server Enforces Certificate Verification

Open the certificate and use the “Certification Path” to view the Intermediate and Root certificates and confirm their respective serial numbers



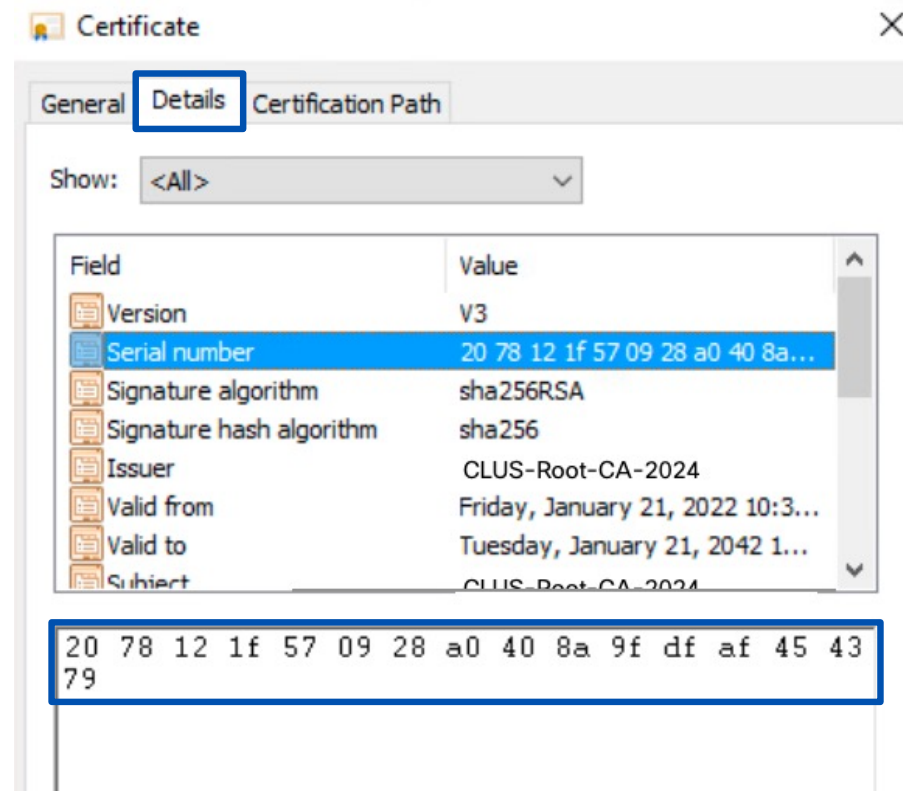
Expressway C Traffic Server Enforces Certificate Verification

Open the certificate and use the “Certification Path” to view the Intermediate and Root certificates and confirm their respective serial numbers



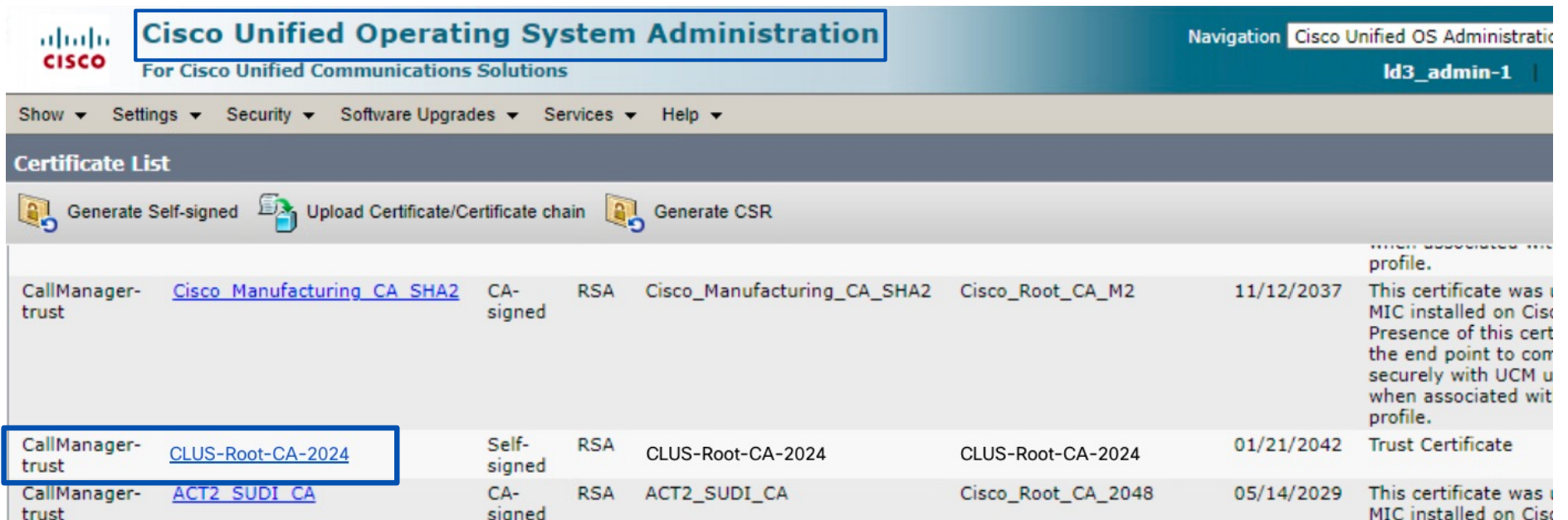
Expressway C Traffic Server Enforces Certificate Verification

Open the certificate and use the “Certification Path” to view the Intermediate and Root certificates and confirm their respective serial numbers



Expressway C Traffic Server Enforces Certificate Verification

Check the CUCM CallManager-trust and tomcat-trust stores to confirm the Expressway C Intermediate and Root CA certificates are installed



The screenshot displays the Cisco Unified Operating System Administration interface. The top navigation bar includes the Cisco logo, the title "Cisco Unified Operating System Administration", and the subtitle "For Cisco Unified Communications Solutions". The right side of the navigation bar shows "Navigation" and "Cisco Unified OS Administration". Below the navigation bar, there is a menu with options: Show, Settings, Security, Software Upgrades, Services, and Help. The main content area is titled "Certificate List". Below this title, there are three icons with labels: "Generate Self-signed", "Upload Certificate/Certificate chain", and "Generate CSR". The "Certificate List" table contains three rows of certificate information. The first row is for "CallManager-trust" with a link to "Cisco_Manufacturing_CA_SHA2", "CA-signed", "RSA", "Cisco_Manufacturing_CA_SHA2", "Cisco_Root_CA_M2", "11/12/2037", and a description. The second row is for "CallManager-trust" with a link to "CLUS-Root-CA-2024", "Self-signed", "RSA", "CLUS-Root-CA-2024", "CLUS-Root-CA-2024", "01/21/2042", and a description. The third row is for "CallManager-trust" with a link to "ACT2_SUDI_CA", "CA-signed", "RSA", "ACT2_SUDI_CA", "Cisco_Root_CA_2048", "05/14/2029", and a description. The second row is highlighted with a blue box.

Trust Store	Link	Signature	Key Size	Issued By	Issued To	Expiration	Description
CallManager-trust	Cisco_Manufacturing_CA_SHA2	CA-signed	RSA	Cisco_Manufacturing_CA_SHA2	Cisco_Root_CA_M2	11/12/2037	This certificate was installed on Cisco Unified Communications Manager. Presence of this certificate at the end point to communicate securely with UCM is required when associated with a profile.
CallManager-trust	CLUS-Root-CA-2024	Self-signed	RSA	CLUS-Root-CA-2024	CLUS-Root-CA-2024	01/21/2042	Trust Certificate
CallManager-trust	ACT2_SUDI_CA	CA-signed	RSA	ACT2_SUDI_CA	Cisco_Root_CA_2048	05/14/2029	This certificate was installed on Cisco Unified Communications Manager.

Expressway C Traffic Server Enforces Certificate Verification

If the Expressway C Intermediate and Root CA certificates are already installed, check the Serial Number of the certificates in CUCM to confirm they are the same as identified from the Expressway C certificate chain

The screenshot displays the 'Certificate Details' page in CUCM for the certificate 'campjetpack-LD3-INT-ADDNS-CA, CallManager-trust'. The page is divided into several sections:

- Actions:** Delete, Download .PEM File, Download .DER File.
- Status:** Status: Ready.
- Certificate Settings:**
 - File Name: CLUS-Root-CA-2024.pem
 - Certificate Purpose: CallManager-trust
 - Certificate Type: trust-certs
 - Certificate Group: product-cm
 - Description(friendly name): Trust Certificate
- Certificate File Data:**
 - Version: V3
 - Serial Number: 2078121F570928A0408A9FDFAF454379 (highlighted with a blue box)
 - SignatureAlgorithm: SHA256withRSA (1.2.840.113549.1.1.11)
 - Issuer Name: CN=CLUS-Root-CA-2024, DC=TACCOL, DC=local
 - Validity From: Fri Jan 21 09:37:42 EST 2022

On the left, the 'Certificate List' sidebar shows 'CallManager-trust' selected, with links to 'Cisco Manager-trust', 'CLUS-Root-CA-2024.pem', and 'ACT2_SUI-CA-2024.pem'. On the right, a partial view of the 'Id3_admin-1' profile is visible.

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Expressway C Traffic Server Enforces Certificate Verification

If the Expressway C CA certificates are uploaded to the CUCM trust stores and the serial numbers have been validated, then the next step would be to restart CallManager, TFTP, and tomcat services

Cisco Unified Serviceability
For Cisco Unified Communications Solutions

Alarm ▾ Trace ▾ Tools ▾ Snmp ▾ CallHome ▾ Help ▾

Control Center - Feature Services

Start Stop **Restart** Refresh Page

Status
Ready

Select Server
Server: CUCMpub.CLUS2024.com --CUCM Voice/Video Go

Performance and Monitoring Services

Service Name	Status	Activation Status
☐ Cisco Serviceability Reporter	Started	Activated
☐ Cisco CallManager SNMP Service	Started	Activated

CM Services

Service Name	Status	Activation Status
☐ Cisco CallManager	Started	Activated
☐ Cisco IP Voice Media Streaming App	Started	Activated
☐ Cisco CTIManager	Started	Activated
☐ Cisco Extension Mobility	Started	Activated
☐ Cisco DHCP Monitor Service	Started	Activated
☐ Cisco Location Bandwidth Manager	Started	Activated
☐ Cisco Directory Number Alias Lookup	Started	Activated
☐ Cisco Headset Service	Started	Activated
☐ Cisco Dialed Number Analyzer Server	Started	Activated
☐ Cisco Dialed Number Analyzer	Started	Activated
☐ Cisco Tftp	Started	Activated

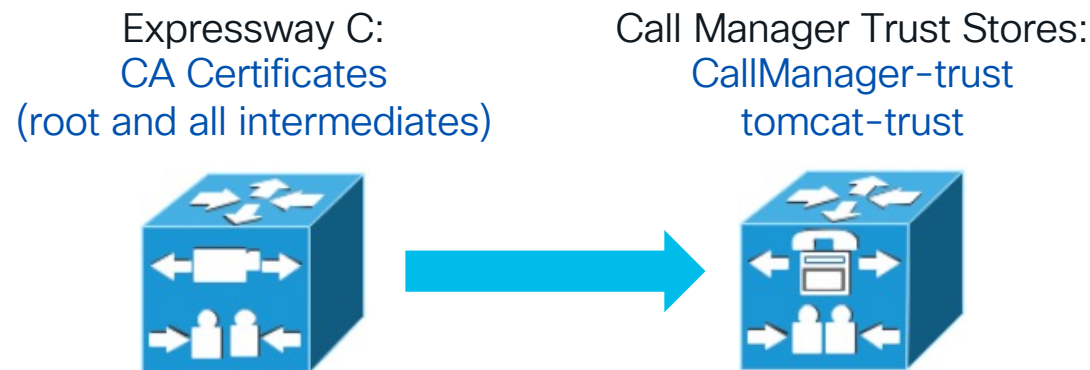
CallManager and TFTP can be restarted from the web UI.

Tomcat can only be restarted from the CLI

```
admin:utils service restart Cisco Tomcat
Do not press Ctrl+C while the service is restarting.
arted properly, execute the same command again.
Service Manager is running
Cisco Tomcat[STOPPING]
Cisco Tomcat[STOPPING]
Cisco Tomcat[STOPPING]
Cisco Tomcat[STOPPING]
Cisco Tomcat[STARTING]
Cisco Tomcat[STARTING]
Cisco Tomcat[STARTED]
admin:
```

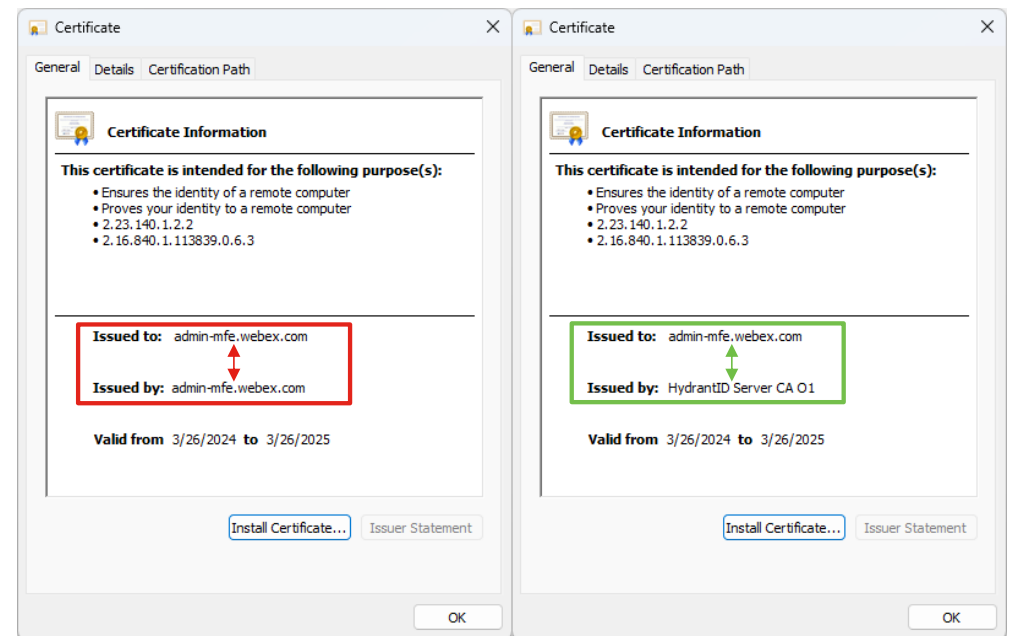
Expressway C Traffic Server Enforces Certificate Verification

This improvement requires the Expressway-C certificate signing CAs (root and intermediates) to be added in CUCM tomcat-trust and CallManager-trust list. After adding these certificates, **you must also restart TFTP, CallManager and Tomcat services on each CUCM node in the cluster.**



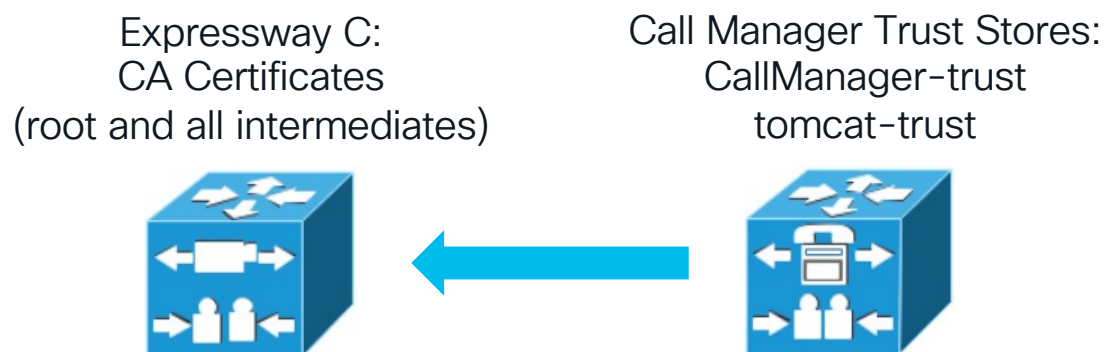
Expressway prevents duplicate self-signed certificates with the same Common Name

- Expressway previously allowed the upload of two self-signed certificates with identical Common Names (CN) to its trusted CA store, despite the underlying restriction preventing the use of two self-signed certificates with the same CN. This behavior could potentially lead to confusion or misconfiguration.
- However, this limitation has been addressed in subsequent versions, including x14.2 and x14.0.7 and upwards. Expressway now prevents the upload of a second self-signed certificate if a certificate with a matching CN already exists in the trusted CA store. This enhancement aims to ensure more consistent and secure certificate management practices.



Expressway C Certificate Validation Change

Starting with version x14.2, Expressway now prioritizes cipher lists differently. This change may lead to unexpected TLS connections following a software upgrade. Prior to the upgrade, it may have been configured to request the Cisco Tomcat or Cisco CallManager RSA certificate from CUCM (or any other product with a separate certificate for the ECDSA algorithm). However, post-upgrade, it might prioritize the ECDSA variant, which is more secure than RSA.



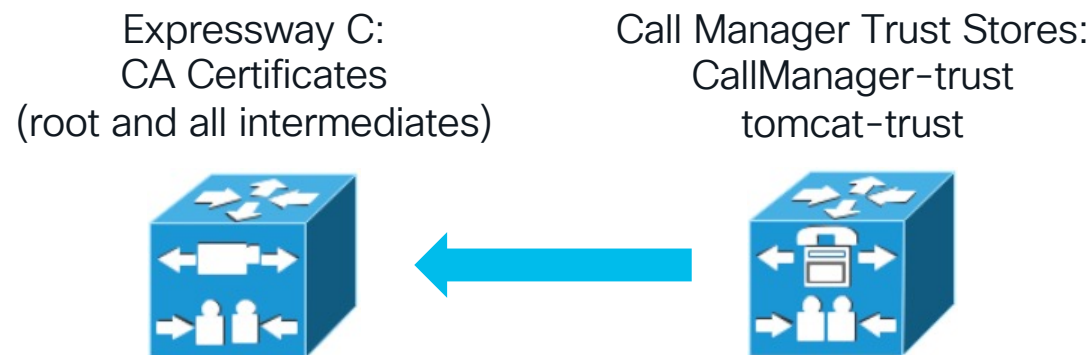
Expressway C Certificate Validation Change

The identifying signature for this issue is Expressway connections to CUCM fails after establishing a TLS connection with errors “[self signed certificate server](#)” and “[502 connect failed](#)”

```
2024-01-05T12:33:14.901-05:00 clus2024expresswayc traffic_server[10907]: [ET_NET 2] WARNING: Core server certificate verification failed for (cucm.clus2024). Action=Terminate Error=self signed certificate server=cucm.clus2024(0.0.0.10) depth=0
2024-01-05T12:33:14.901-05:00 clus2024expresswayc traffic_server[10907]: [ET_NET 2] ERROR: SSL connection failed for 'cucm.clus2024': error:1416F086:SSL routines:tls_process_server_certificate:certificate verify failed
2024-01-05T12:33:14.902-05:00 clus2024expresswayc traffic_server[10907]: UTCTime="2024-01-05 17:33:14,902" Module="network.http.trafficserver" Level="INFO": Detail="Sending Response" Txn-id="56" TrackingID="" Dst-ip="127.0.0.1" Dst-port="32790" Msg="HTTP/1.1 502 connect failed"
2024-01-05T12:33:14.902-05:00 clus2024expresswayc traffic_server[10907]: UTCTime="2024-01-05 17:33:14,902" Module="network.http.trafficserver" Level="DEBUG": Detail="Sending Response" Txn-id="56" TrackingID="" Dst-ip="127.0.0.1" Dst-port="32790"
HTTPMSG:
|HTTP/1.1 502 connect failed
Date: Fri, 05 Jan 2024 17:33:14 GMT
Connection: keep-alive
Server: CE_C
Cache-Control: no-store
Content-Type: text/html
Content-Language: en
Content-Length: 247
```

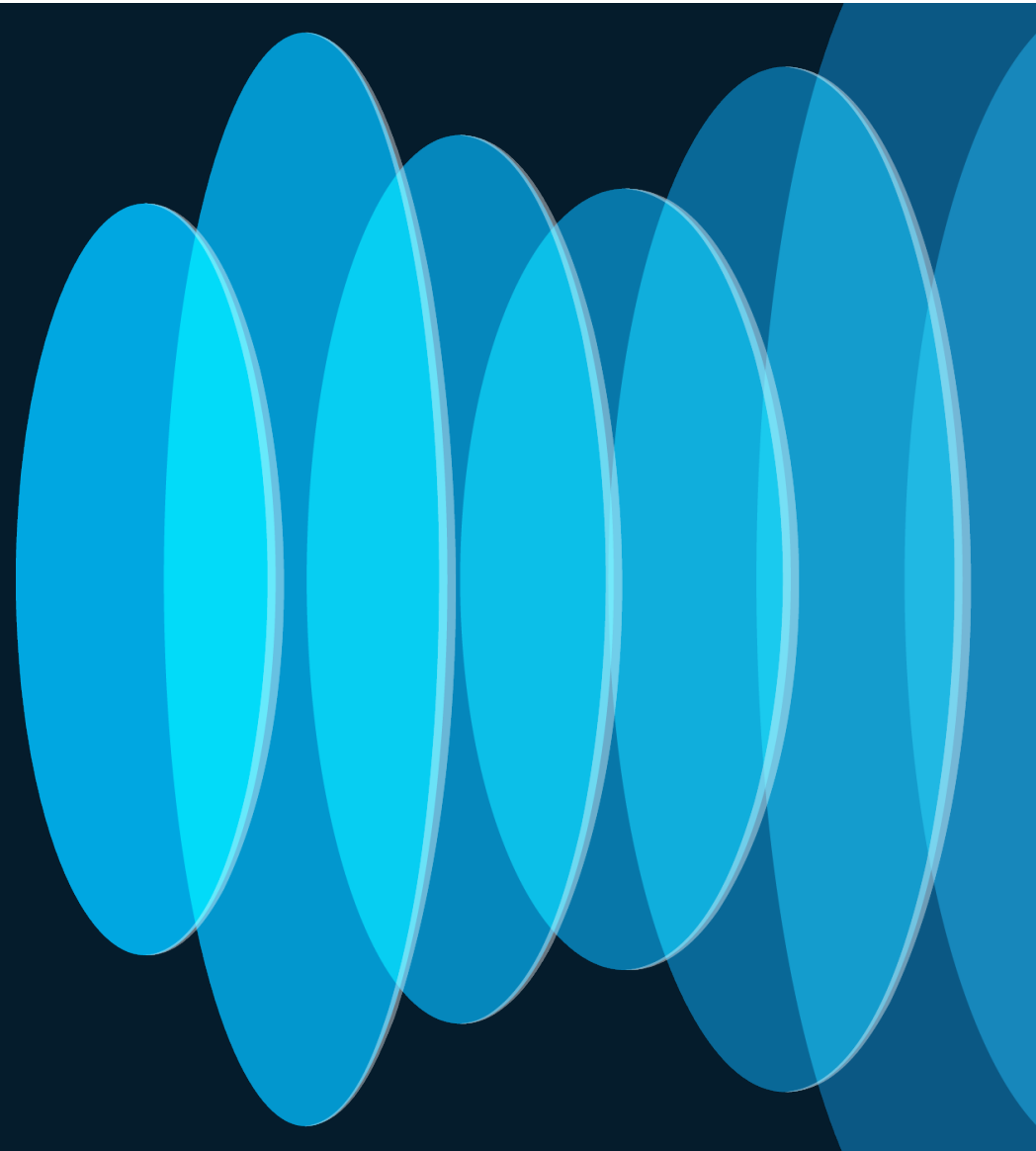
Expressway C Certificate Validation Change

Cisco Tomcat-ECDSA and Cisco CallManager-ECDSA certificates need to be signed by a distinct CA. Uploading self-signed certificates to the Expressway-C Trusted CA database isn't feasible due to identical common names.



Upcoming!

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Cisco Support Assistant – Certificates Walk Through



Cisco Support Assistant

Knowledge Scope: Collaboration – Troubleshooting Guided Experience



It appears you need assistance with Expressway Certificates, let me help you with that.

We've put together this quick guide that can walk you through generating or renewing your Expressway certificates.

First, we'll ask a few questions to help determine your needs and then we'll provide step by step instructions to walk you through requesting and installing your certificates.



How can we assist you with your Expressway Certificates?

- ☐ I need to generate or renew my Expressway certificates.
- ☐ I need documentation on Expressway certificate requirements.
- ☐ I need help selecting a Certificate Authority.
- ☐ My problem is not listed.



Please select the services that you will be using in this deployment (select all that apply)

☐ Mobile and Remote Access ☐ Business to Business ☐ Cisco Meeting Server WebApp ☐ XMPP Federation

Cisco Support Assistant – Certificates Walk Through

Greetings,

Please find attached the PDF file (*jgamboag-20240524-140249.pdf*) including the results of the System Health Check generated during your recent Guided Experience session.

Best regards,

-Collaboration TAC Automation team

Questions or feedback? Email us here: tac-collab-automation-feedback@cisco.com

Email generated with
PDF attached

PDF with Contents

Contents

Introduction

Generating CSR for Expressway-E

Updating Trust-List on Expressway-E for Expressway-E certificate

Updating Trust-List on Expressway-C for Expressway-E certificate

Installing Certificate for Expressway-E

Conclusion

A banner for Cisco Live! featuring a dark blue background with stylized, overlapping hills in various shades of green and blue at the bottom. The text "Continue your education" is written in white, sans-serif font.

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How to troubleshoot 9800-AireOS Mobility - TACENT-2018

Ivan Zamudio Zamudio Gomez, Technical Consulting engineer, Cisco Systems, Inc.
Katy Torres Romo, Technical Consulting Engineer, Cisco Systems, Inc.

[Schedule](#)

Thursday, Jun 6 | 10:45 am - 11:15 am PDT

[EXPLORER](#) [FULL CONFERENCE](#) [IT LEADERSHIP](#) [FULL CONFERENCE PLUS](#)

Nexus Dashboard/NDFC Fabric Discovery Troubleshooting - TACDCN-2009

Jeffrey Mertes, Technical Consulting Engineer, Cisco Systems, Inc.

[Schedule](#)

Thursday, Jun 6 | 11:30 am - 12:00 pm PDT

[EXPLORER](#) [FULL CONFERENCE](#) [IT LEADERSHIP](#) [FULL CONFERENCE PLUS](#)

TCL & EEM Scripts and Guest Shell - TACENT-2015

Stan Fofano, Technical Consulting Engineer, Cisco Systems, Inc.

[Schedule](#)

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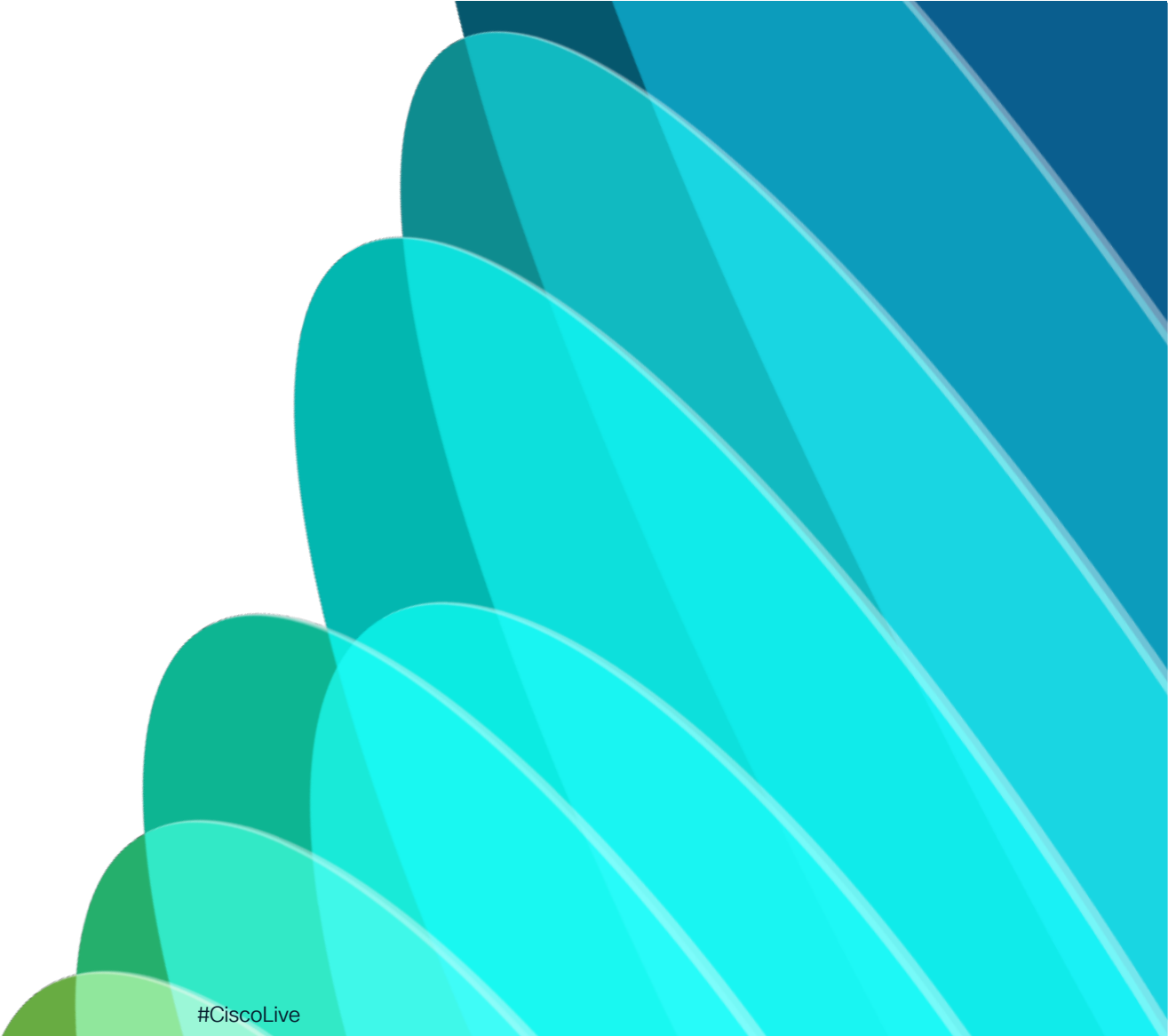


Complete your surveys in the Cisco Live mobile app.

Thank you

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Q & A

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